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Clinical Manual of Pediatric Nursing

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Contents

Communication.....5
What is communication? 6
What are the types of Communication? 6
1- Verbal communication: 6
2-NONVERBAL COMMUNICATION 6
Pediatric Health History.....10
The Health History 11
Types of Pediatric Health Histories.....11
1. Initial (comprehensive) health history.....11
2. Interval (well-child) health history.....11
3. Focused (episodic) health history.....11
4. Follow-up health history 11
5. Emergency health history.....11
DATA COLLECTION 12
1. Subjective data. 12
2. Objective data.....12
Methods of Data Collection.....12
How to Conduct a Pediatric History Interview 12
Infants and Young Children 13
School-Aged Children and Adolescents.....13
Age-specific interview and assessment tips. 13
Infant.....13
Toddler.....14
Preschool child 14
School-age child 15
Adolescent 15
Review of systems: 16
Physical examination 21
There are two types of physical examination: 22
B.Focused physical examination 22

EXAMINATION TECHNIQUES.....22
PREPARATION OF THE CHILD.....23
Preparation of the Environment.....23
Sequence of Pediatric Physical Examination According to Age and Developmental Level.....24
PHYSICAL ASSESSMENT OF THE NEWBORN.....26
Anthropometric measurements.....30
Anthropometry.....31
Weight measured:.....31
Weight progression (1-12 month).....31
Height or Length.....31
Head circumference:.....33
Chest circumference: 34
Body Mass Index 35
Physiologic measurements.....36
Temperature.....37
Respiration.....40
Blood Pressure 41
Reflexes.....42
Primitive reflexes.....43
1.Rooting reflex:….....43
2.Sucking reflex:….....43
3.Startle reflexes:….....43
4.The tonic neck reflex. 43
5.The Moro reflex:.....43
6.The grasping reflex.....43
7.The Babinski reflex: 43
8.Dancing (or stepping) reflex:.....43
9.Palmar grasp:… 43
10.Plantar grasp:.. 43
11.Protective reflexes: 43

Fluid Calculation and Calories Requirement.....84
Body fluid composition 85
1. Intracellular 85
2. Extracellular.....85
Calculating Pediatric Fluid Needs 85
Fluid needs based on weight.....85
The hourly maintenance requirement can be calculated from the following formula: 86
Types of I.V. solutions 86
Energy / Calories requirement 87
There are three type of milk each cup (200ml) contain: 88
Calculating a Formula's Adequacy 89
Oxygen Therapy.....90
Introduction: 90
Definition:.....91
Purpose: 91
Indication: 91
Types:.....91
Methods: 92
1- Simple face mask 92
2- Nasal Cannula 92
3- Head box (hood).....93
4- Mist tent.....93
5- Incubator.....93
Equipment:.....94
Humidification.....94
Procedures of nasal cannula, mask or hood.....94
Immunization and vaccination.....97
Immunization.....97
Nasogastric Intubation 106
Definition of Gavage feeding: 107

Indications: 107
Purpose of gavage feeding:.....107
Guidelines of nasogastric tube selection: 107
Type of tube:.....108
Tube size:.....108
Maximize the Feeding Pleasure of the Infant.....110

Communication

Learning objectives:

Upon completion of this class, the student will be able to:

1-Explain what communication is and its importance in developing positive relationships with children and their families.

2-Identify communication strategies for interviewing parents.

3- Formulate guidelines for communicating with child

What is communication?

Communication is defined as the exchange of meanings between and among individuals through a shared system of symbols. The sender, message, channel, receiver, and feedback are major components of the communication process Communication is both talking and listening.

What are the types of Communication?

1- Verbal communication:

Verbal communication refers to messages that are communicated through words and language (Estes, 2010). It is most effective if it is brief, clear, effectively toned, paced appropriately, relevant, and well timed

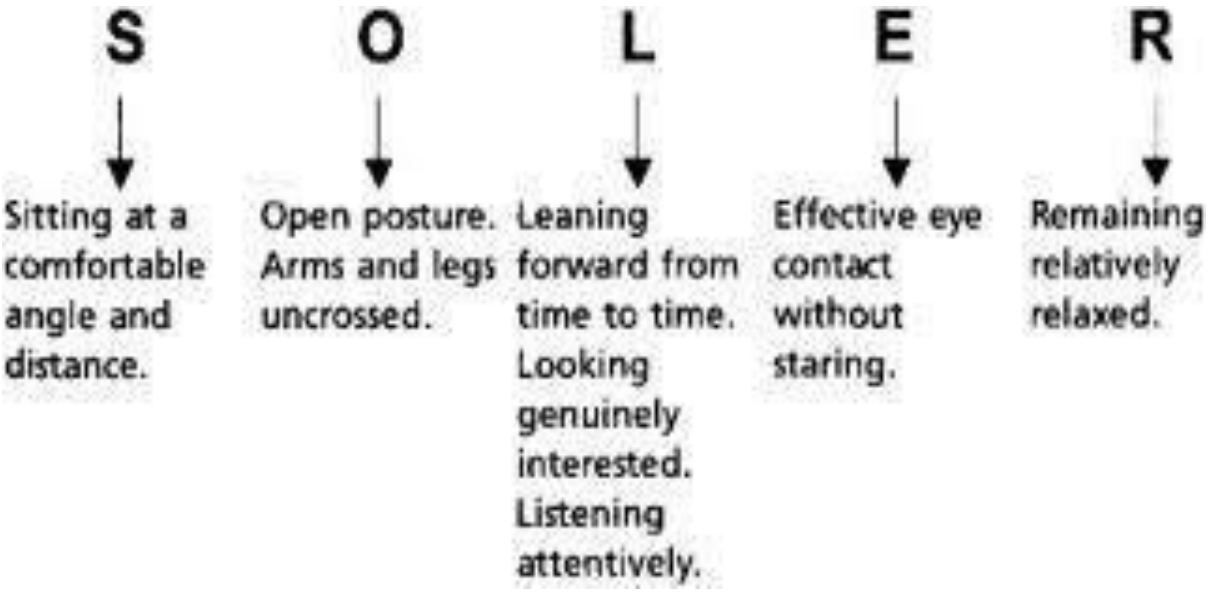
Important points should be repeated and medical jargon avoided, especially when talking to children and their caregivers and families.

*Messages also need appropriate pacing; they should not be too slow or fast, and there should be few pauses and periods of silence, for example, teaching will not be effective if delivered when children are in pain or have visitors, or when caregivers are preoccupied with personal thoughts. When interacting with families, it is also essential to convey interest and warmth; to avoid distractions, yes or no questions, and personal bias; and not monopolize the conversation.

2-NONVERBAL COMMUNICATION

Nonverbal communication conveys feelings, attitudes, and intentions. In fact, most communication is nonverbal, and most verbal communication contradicts nonverbal messages. Listeners will believe the nonverbal message rather than the verbal message if they contradict each other. Therefore, it is critical to always consider nonverbal communication in any interaction since it contributes to what others “hear.” The nurse should use language consistent with the child’s developmental level and deliver the message so the language is consistent with the nonverbal cues. For example, never smile or laugh when giving care that may be frightening or painful. Avoid words that have violent or scary connotations, such as stick, shot, and sting

Nonverbal communication includes: spatial relationships (the distance between participants); appearance (clothing, grooming, hair style); eye contact; body posture (slouching, standing erect with the head leaning to receiver); gestures (which sometimes add emphasis to words); facial expressions (which need to agree with the message); timing (hesitation may imply untruthfulness or diminish the effect of the message); and, with children especially, touch.



DEVELOPMENTAL LEVEL	COMMUNICATION PRINCIPLES
Infants	Allow the infant time to warm up to strangers. Respond to the infant's cries in a timely manner. Use motherese, and a soothing and calm voice. Talk to the infant directly.

Toddlers	Approach at eye level if possible. Toddlers Approach the toddler carefully; the child may be fearful. Integrate the toddler's words for familiar objects or activities into care. Prepare for procedures right before they are to be carried out. Integrate dolls, storytelling, and picture books into conversations. Stand at eye level.
Pre-schoolers	Allow choices as appropriate. Use play, storytelling, puppets, and third parties. Speak honestly, use simple language, and be concise. Prepare for procedures 1 hour or less before they are to be experienced.
School-age children	Use books, diagrams, and videos in preparing for procedures. Prepare for procedures several days in advance. Allow for honest expression of feelings and adequate time for questions to be answered.

Adolescents	Prepare for up to 1 week prior to experiences. Respect the adolescent's need for privacy. Use appropriate medical terminology. Use creative methods to explain experiences and procedures.Respect the adolescent's right to refuse treatment.

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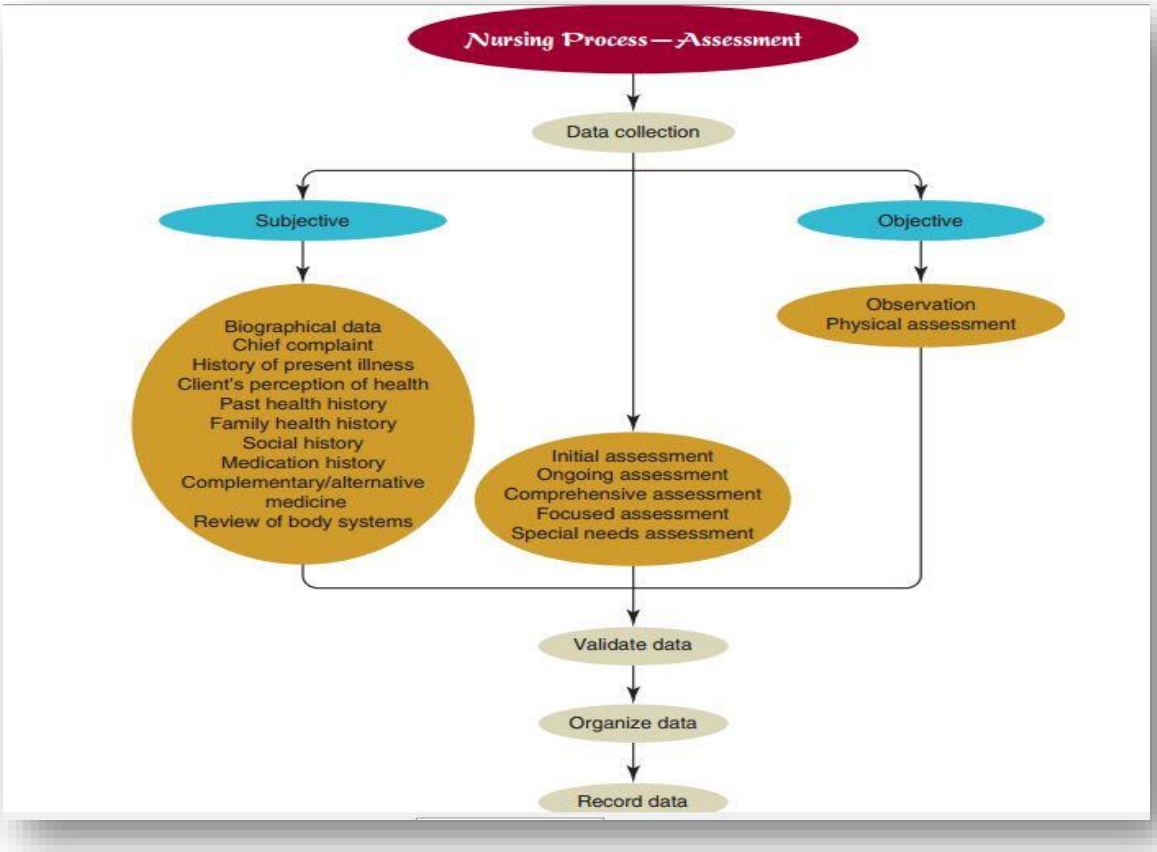
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Pediatric Health History

Objective

1. Define health history
2. Numerate and describe types of Pediatric Health Histories
3. Differentiate between objective and subjective data.
4. Describe Methods of Data Collection.
5. Identify How to Conduct a Pediatric History Interview.



The Health History

A health history is a collection of **subjective data** that provides a detailed profile of the patient's health status. Nurses use therapeutic communication skills and interviewing techniques during the health history to establish an effective nurse–patient relationship and to gather data to identify actual and potential health problems.

Types of Pediatric Health Histories (Chiocca, 2015)

1. Initial (comprehensive) health history**:**

   An initial health care visit requires a **comprehensive health history**, which creates a database of information for the child who is new to the provider's practice. The data collected in this history focus not just on past or current illnesses, but also on wellness, health patterns, health knowledge, and activities of daily living (ADL). The content of this history creates a baseline against which all future changes in the child's health status are measured.

2. Interval (well-child) health history**:**

   Subsequent well-child visits require an interval history; that is, a history that updates the patient's health status since the last office visit.

3. Focused (episodic) health history**:**

   This history is problem-centered, focusing on the present illness or problem. A focused health history is all that is necessary when the child presents with an
   acute problem or the exacerbation of a chronic condition.

4. Follow-up health history**:**

   When the child returns to the office after an illness or injury, a follow-up history is taken to assess whether the problem is the same, better, or worse.

5. Emergency health history**:**

   An emergency health history involves the quick collection of data in an acute situation. While this history resembles a focused history, it needs to be obtained rapidly, usually simultaneously with

interventions, such as the administration of medications or respiratory treatments.

DATA COLLECTION (Taylor, Lillis, LeMone, & Lynn, 2011)

Two Types Of Data: Subjective and Objective.

1. Subjective data are information perceived only by the affected person; these data cannot be perceived or verified by another person. Examples of subjective data are feeling nervous, nauseated, or chilly and experiencing pain. Subjective data also are called **symptoms or covert data**.
2. Objective data are observable and measurable data that can be seen, heard, or felt by someone other than the person experiencing them. Objective data observed by one person can be verified by another person observing the same patient. Examples of objective data are an elevated temperature reading (e.g., 101F), skin that is moist, and refusal to look at or eat food. Objective data also are called **signs** or **overt data**.

Methods of Data Collection

1-Observation is the conscious and deliberate use of the five senses to gather data (objective data).

2-Patient interview the nurse obtains a nursing history by interviewing the patient. (An **interview** is a planned communication)

3-Physical assessment (or *physical examination*) produces primarily **objective data** and makes use of the techniques of inspection (visual examination), palpation (touch), percussion (tapping a body surface), direct auscultation (listening with the unaided ear), and indirect auscultation (listening with a stethoscope).

How to Conduct a Pediatric History Interview (Chiocca, 2015)

Obtaining an accurate and complete pediatric health history is essential when providing health care to children. The pediatric health care provider must possess excellent **communication skills** when working with children and families in order to elicit the most accurate and complete health history while maintaining provider– patient trust.

Infants and Young Children

• For **infants** and preverbal children, the **primary source** of historical information is the parent or primary caregiver.

For older infants and **toddlers,** it is important that the parent remain within eyesight of the child to minimize stranger anxiety. Parents can then concentrate on answering questions rather than tending to a crying child.

Some **preschoolers** can answer questions during the history, depending on their language abilities, developmental level, and temperament. These questions may involve play, toys, likes, dislikes, friends, and school.

Young children who are very shy should not be forced to participate in the interview, and children should not be overwhelmed with too many questions. The health care provider should remember to use simple terms and avoid medical jargon.

School-Aged Children and Adolescents

• Children who are **school-aged and older** can provide their own health history with verification from the parent or caregiver as needed. Beginning the encounter with the parent in the room is important to make the child feel comfortable and secure.

Older school-aged children and adolescents have a heightened need for privacy. For this reason, the parent should be asked to leave the room for certain portions of the history, such as questions that deal with drug or alcohol use, sexual activity, depression, and risk-taking behaviors. The **preteen or adolescent** is more likely to give accurate answers without the parent present.

Age-specific interview and assessment tips. (Lippincott, 2012)

Infant

• Before performing a procedure, talk to and touch the infant.
• Use a gentle touch.
• Speak softly.
• Allow the infant to hold a favorite toy during the assessment.

• Let an older infant hold a small block in each hand.
• Remember that an older infant may be wary of strangers.
• Be alert to infant cues, such as crying, kicking, or waving arms.
• Perform traumatic procedures last when the infant is crying.
• Use distractions, such as bright objects, rattles, and talking.
• Enlist the parent's aid when examining the ears and mouth.
• Avoid abrupt, jerky movements.
• When the child is quiet, auscultate the heart, lungs, and abdomen.

Toddler

• Encourage the parents to be with you during the interview.
• Allow the toddler to be close to his parents.
• Provide simple explanations and use simple language.
• Use play as a communication tool.
• Tell the toddler that it's okay to cry.
• Watch for separation anxiety.
• Use the toddler's favorite toy as a tool during the interview. Encourage the toddler to use the toy for communication.
• Use play (count fingers or tickle toes) to assess body parts.
• Use parent assistance during the examination. For example, ask the parents to remove the toddler's outer clothing and help restrain the child during eye and ear examination.
• Use encouraging words during the examination.

Preschool child

• Ask simple questions.
• Allow the child to ask questions.
• Provide simple explanations.
• Avoid using words that sound threatening or have double meanings.
• Avoid slang words.
• Validate the child's perception.
• Use toys for expression.
• Use simple visual aids.

• Enlist the child's help during the examination, such as by allowing him to give you the stethoscope.
• Allow the child to touch and operate the diagnostic equipment.

• Explain what the child is going to feel before it happens. For example, explain that the stethoscope will be cold before using it on the child.
• Utilize the child's imagination through puppets and play.
• Give the child choices when possible.

School-age child

• Provide explanations for procedures.
• Explain the purpose of equipment, such as an ophthalmoscope to see inside the eye.
• Avoid abstract explanations.
• Help the child vocalize his needs.
• Allow the child to engage in the conversation.
• Perform demonstration.
• Allow the child to undress himself.
• Respect the child's need for privacy.

Adolescent

• Give the adolescent control whenever possible.
• Facilitate trust and stress confidentiality.
• Encourage honest and open communication.
• Be nonjudgmental.
• Use clear explanations.
• Ask open-ended questions.
• Anticipate that the adolescent may be angry or upset.
• Ask if you can speak to the adolescent without the parent present.
• Ask the adolescent about parental involvement before initiating it.
• Give your undivided attention to the adolescent.
• Respect the adolescent's views, feeling, and differences.
• Allow the adolescent to undress in private, and provide the child with a gown.
• Expose only the area to be examined.
• Explain findings during the examination.
• Emphasize the normalcy of adolescent's development.
• Examine genitalia last but examine them as you would examine any other body part.

BOX 4-4 OUTLINE OF A PEDIATRIC HEALTH HISTORY

Identifying information
1. Name
2. Address
3. Telephone
4. Birth date and place
5. Race or ethnic group
6. Sex
7. Religion
8. Date of interview
9. Informant

Chief complaint (CC)—To establish the major specific reason for the child's and parents' seeking professional health attention

Present illness (PI)—To obtain all details related to the chief complaint

Past history (PH)—To elicit a profile of the child's previous illnesses, injuries, or operations
1. Birth history (pregnancy, labor and delivery, perinatal history)
2. Previous illnesses, injuries, or operations
3. Allergies
4. Current medications
5. Immunizations
6. Growth and development
7. Habits

Review of systems (ROS)—To elicit information concerning any potential health problem
1. General
2. Integument
3. Head
4. Eyes
5. Ears
6. Nose
7. Mouth
8. Throat
9. Neck
10. Chest
11. Respiratory
12. Cardiovascular
13. Gastrointestinal
14. Genitourinary
15. Gynecologic
16. Musculoskeletal
17. Neurologic
18. Endocrine

Family medical history—To identify genetic traits or diseases that have familial tendencies and to assess exposure to a communicable disease in a family member and family habits that may affect the child's health, such as smoking and chemical use

Psychosocial history—To elicit information about the child's self-concept

Sexual history—To elicit information concerning the child's sexual concerns or activities and any pertinent data regarding adults' sexual activity that influences the child

Family history—To develop an understanding of the child as an individual and as a member of a family and a community
1. Family composition
2. Home and community environment
3. Occupation and education of family members
4. Cultural and religious traditions
5. Family function and relationships

Nutritional assessment—To elicit information on the adequacy of the child's nutritional intake and needs
1. Dietary intake
2. Clinical examination

Review of systems:

General

Overall state of health; recent weight loss or gain (compare to growth charts); obesity; delays in physical, psychosocial, or language development; fatigue; weakness; malaise; difficulty performing activities of daily living; or chronic pain.

Integument
General skin texture, birthmarks, or pigmentation; skin color changes (jaundice, cyanosis, mottling, pallor, bruising, or petechiae); rashes or lesions (location and description); excessive skin dryness; pruritus; hives; atopic dermatitis; contact dermatitis; seborrhea; acne; changes in moles; changes in body hair (excessive hair growth or loss or hair color change); nail disorders; or pediculosis.

Head
Head size and shape (including fontanels), microcephaly, macrocephaly, head injuries, headaches, or dizziness.

Eyes
Visual problems (blurred vision, double vision, bumping into objects, sitting very close to the television or computer screen, inability to see blackboard, holding book very close to face, squinting, rubbing eyes, photophobia, use of glasses or contact lenses), date of last eye exam and results, strabismus, dizziness, eye drainage or infections, excessive tearing, or eyelid edema.

Ears
Ear infections (e.g., otitis media, otitis externa), ear pain, ear drainage, surgery (e.g., myringotomy with tympanostomy tubes), delayed speech, evidence of hearing loss, date of last hearing exam and result, or tinnitus.

Nose
Frequent rhinorrhea, upper respiratory infections, sinusitis, allergic rhinitis, stuffy nose, nasal congestion, allergies, mouth breathing, snoring, obstructive sleep apnea, epistaxis, allergic salute, allergic crease, dark circles under eyes, or altered sense of smell.

Mouth/throat
Pattern of tooth eruption or loss, number of teeth, teething (infants), dental care, use of fluoride, dental caries, age of first dental visit, date of last visit to dentist, frequency of dental visits, tooth pain, history of dental trauma, sores in mouth or tongue, history of oral candida infections, gum bleeding,

mouth breathing, tonsillar enlargement or infections, history of tonsillectomy, postnasal drip, cleft lip or palate, sore throats, streptococcal infections, difficulty chewing or swallowing, hoarseness, or voice and cry irregularity.

Chest

Pain, newborn breast enlargement, male gynecomastia, breast lesions, discharge, or enlarged

Neck

Limitation of movement, pain, stiffness, torticollis, thyroid enlargement, or lymphadenopathy axillary nodes.

Lymphatics

Frequent infections, fevers, pain, and swelling, tenderness of any lymph nodes, hepatosplenomegaly, chills, night sweats, or pallor.

Respiratory

Frequent respiratory infections, dyspnea (shortness of breath), apnea, stridor, croup, pneumonia, cystic fibrosis, asthma, wheezing, chronic cough, sputum production, reduced exercise tolerance, dyspnea at rest or on exertion, cyanosis, or tuberculosis; date of last chest radiograph and skin reaction from tuberculosis skin testing; or history of smoking or secondhand smoke exposure.

Cardiovascular

Murmurs, congenital heart defects, poor infant feeding, slow or decreased weight gain, chest pain, cyanosis, tachypnea, tachycardia, hypertension, edema, cold extremities, dizziness, palpitations, fainting spells, exercise intolerance, dyspnea on exertion, fatigue, diaphoresis, rheumatic fever, anemia, or recent blood transfusion.

Gastrointestinal

Appetite, food intolerances, dietary history, abdominal pain, ulcers, nausea, vomiting, reflux, recent changes in bowel patterns, gastrointestinal infections, diarrhea, belching, flatulence, constipation, type and frequency of stools, use of laxatives, blood in stools, jaundice, anal itching, fissures, or pinworms.

Genitourinary

Dysuria, frequency, urgency, burning, hesitancy, enuresis, urinary tract infections, discharge, polyuria, oliguria, hematuria, character of stream, or flank pain; history of epispadias or hypospadias and date of surgery; history of cryptorchidism (include age of child and if orchiopexy performed, include date of surgery); hernia, hydrocele, swelling of scrotum when crying, performance of testicular self-exam in adolescent males, sexually transmitted infections, or sexual activity.

Gynecologic

Age of menarche, date of last menstrual period, length of menses, frequency of cycle, dysmenorrhea, history of heavy menstrual bleeding, vaginal discharge, abnormal vaginal bleeding, sexual activity, type of contraception, date of last Pap smear, or obstetric history, if applicable

Musculoskeletal

Usual activity level, history of weakness, clumsiness, ataxia, lack of coordination, unusual movements, joint or muscle pain, cramps or inflammation; swelling, sprains, fractures, mobility problems, back or joint stiffness, abnormal gait, limps, or spinal curvatures.

Neurologic

General affect and mood, ataxia, tremors, tics, dizziness, syncope, tingling, sensory changes, paresthesia, unusual movements, seizures, loss of consciousness, memory loss, headaches, aphasia or other speech problems, fears, nightmares, or unusual habits.

Endocrine

Disturbances in growth, polyuria, polyphagia, polydipsia, thyroid disease, obesity, type 1 or 2 diabetes, excessive sweating, salty taste to skin,

intolerance to temperature changes, signs of early or delayed puberty, or abnormal hair distribution.

Hematologic

Anemia, pallor, lymphadenopathy, bleeding disorders, bruising, petechiae, bleeding gums, blood transfusions, toxic drug exposure, radiation, or chemotherapy.

Psychosocial development

Developmental delays, behavior changes, tantrums, breath-holding spells, bedwetting, school failure, social withdrawal, irritability, sleep pattern disturbances, depression, substance abuse, eating disorders, or psychiatric disorders. (Hockenberry & Wilson, 2015)

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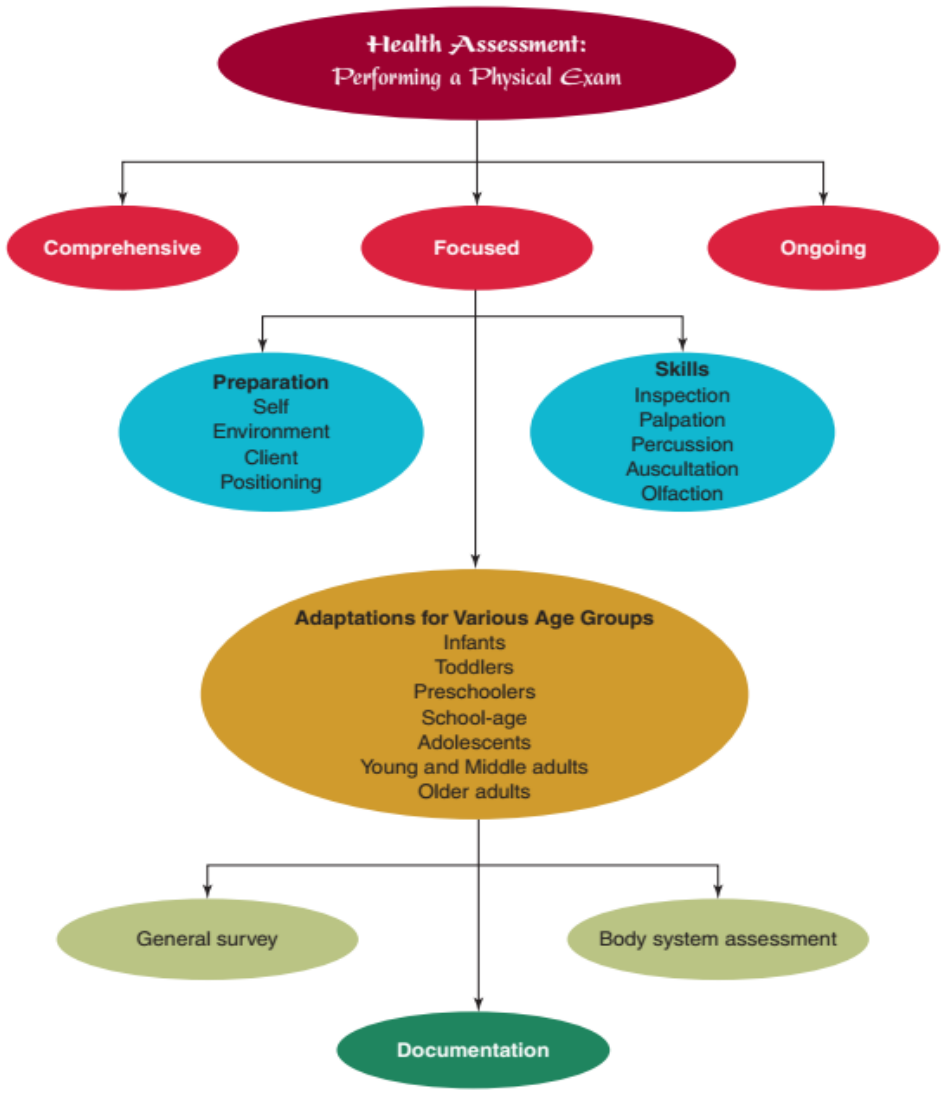
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Physical examination

Objectives:

1. Define physical examination
2. Illustrate type of physical examination.
3. Explain the Preparation of the Environment (examination room).
4. Identify the Sequence of Pediatrics' Physical Examination.



The physical examination is an essential part of the pediatric health assessment. The health care provider must possess strong physical assessment **skills and a knowledge** base specific to the developing bodies and minds of children. These assessment **skills include inspection, palpation, percussion, and auscultation.**

The pediatric physical examination differs from the adult examination in that the approach to the patient and the sequence of the examination differ according to age and developmental level. (Chiocca, 2015)

There are two types of physical examination:

Complete* (head-to-toe) and *focused physical examination

A. Children who are new to the provider's practice or who are being seen for their annual health maintenance visits, school physicals, or sports physicals require a full **head-to-toe physical examination**.
B. **Focused physical examination** complements the focused history and simply concentrates on the area of the chief complaint. This type of physical examination is done when a child presents to the emergency room or clinic with an illness, injury, or other specific complaint.

EXAMINATION TECHNIQUES

Physical assessment involves four major assessment skills:

1. Inspection
2. Palpation
3. Percussion
4. And auscultation.

These skills are normally performed in this order. However, when assessing the abdomen, the sequence is altered: inspection, auscultation, palpation, and then percussion.

When examining a child, providers use their senses to gather objective data. For example, sight is used to assess color, respiratory effort, posture, and gait. Smell assesses hygiene and can help diagnose infections. Touch can help providers find masses or detect skin changes. Hearing allows the provider to assess body sounds.

PREPARATION OF THE CHILD

Before examining the pediatric patients, the provider should first consider the child's age and developmental level. Before beginning the examination, he or she should take time to get acquainted with the child in an age-appropriate manner so that the child is as comfortable as possible. This may involve using play techniques with infants and young children or talking with a school-aged child or adolescent about his or her interests. When appropriate, toddlers, preschoolers, and school-aged children should be allowed to handle the equipment before the examination begins to demonstrate to the child that the equipment will not be painful; for example, allowing the child to listen to heart sounds of the parents or provider (Figure 8.1).



FIGURE 8.1. Allowing children to handle examination instruments may decrease their anxiety and fear.

Preparation of the Environment

The examination room should have a comfortable temperature and be well lit. The room should be in a nonthreatening area and decorated in an age appropriate manner. All instruments should be kept out of the child's view until use, and the examination room should be childproofed. Age-appropriate toys and reading material should also be nearby as a distraction and as a means of assessing development. Privacy should be maintained at all times for children of all ages.

Sequence of Pediatric Physical Examination According to Age and Developmental Level

Age group	Preparation	development Considerations	Sequence
Newborns and young infants (0–6 months)	**Position:** Place supine on exam table. Offer breast /bottle/ pacifier to quiet newborn for exam. Examine while dressed in a diaper only; remove while examining genitourinary area. When examining neonates, use radiant heat warmer to conserve body heat.	Lack of head control until 4–6 months. Infants respond well to smiling faces and soft voices. May respond well to noisemakers or toys to see or touch. Abrupt, rough movements should be avoided.	Perform least distressing aspects of exam first (i.e., if quiet، auscultate heart and lungs; proceed head to-toe, then examine ears, nose, and throat last while the child is crying. Elicit Moro reflex last.
Older infants (6–12 months	**Position:** May need to examine in parent's lap because of **stranger anxiety** if no distress, place on exam table with parent in view. Keep primary caregiver and security object within eyesight of infant. Ensure that exam room is a comfortable temperature. Offer breast/bottle/ pacifier to quiet crying infant. Examine while child is dressed in a diaper only; remove while examining genitourinary area.	Infants prefer smiling faces and soft voices. May respond well to noisemakers or toys to see or hold. Often cry when cold; this may interfere with exam. Abrupt, rough movements should be avoided. **Stranger anxiety** begins at approximately 7–8 months of age. Gather as much objective data as possible by just observing and not touching (infant will often begin crying vigorously when touched by a stranger.(Increased motor abilities cause need for safety measures in exam room; this continues with older pediatric patients, as well.	Perform least distressing aspects of exam first (i.e., if quiet، auscultate heart and lungs; proceed head to toe, then examine ears, nose, and throat last while child is crying).
Toddlers	**Position:** Examine child on parent's lap or exam table, depending on where child is most cooperative. Gain toddler's confidence before beginning exam؛ let	Stranger anxiety decreases by approximately 18 months of age؛ separation anxiety by approximately 2 years. Toddlers are negativistic; often respond "no" when they mean "yes". Dislike body boundary intrusion and often dislike having clothes removed; therefore have parent remove clothes. Offer simple	Perform least distressing and least intrusive aspects of exam first (i.e، auscultate heart and lungs; examine ears،

	toddler see and touch examination instruments; keep parent and security object nearby.	choices but provider should not give toddler a choice where none exists (e.g., "May I look in your ears.("؟ Toddlers are often uncooperative with exam; complete exam as quickly but as thoroughly as possible. May need to enlist parents' help in gently restraining child during exam. Use distraction techniques (play، puppets, toys). Praise toddler for cooperation.	nose, and throat last). Auscultate, palpate، and percuss whenever quiet. Praise toddler for cooperating with exam.
Preschooler	**Position**: <u>Exam table with parent nearby.</u> To allay fears, may use dolls, stuffed animals، or parent to "examine" first before child. Allow preschooler to see and touch equipment. Give preschooler gown; ask child to remove all clothing but underwear.	They like to "help" and are concrete, literal thinkers, and protect their modesty. Magical thinking causes many fears. Provider should take care regarding choice of words (e.g ،.take" your blood pressure.(Preschooler may think she or he is ill because of bad behavior (magical thinking). Preschoolers need feedback during exam Your lungs sound just like they are supposed to.") Games, dolls, or toys sometimes helpful when conducting physical exam to gain child's cooperation. Preschoolers should be given choices when possible	Head to toe if cooperative. Otherwise use approach as with toddler. Should be able to examine older preschoolers head to toe.
School-aged children	Position: <u>Exam table</u> Give child gown and ask child to undress except for underpants. Explain purpose of all equipment. Young school-aged child often prefers parent to be present; older school aged child may prefer parent to leave room during exam.	Can now address questions more directly to child. Answer child's questions honestly and concretely; diagram and teaching dolls are helpful during explanations. Child often sees physical exam as "fun"; likes to learn about body. Protect privacy and modesty	Preform head to toe with genitalia last

Age group	Preparation	development Considerations	Sequence
Adolescents	Position: Exam table Provider should explain confidentiality parameters. Offer option for parent to be present or to leave the room during physical exam. Give teen a gown; allow to undress in private; may leave undergarments on.	Elicit active participation from teen when obtaining health history. Confidentiality and privacy are important. Protect modesty. Address teen directly during exam. Private Discuss issues of sexuality and body development; emphasize normalcy whenever possible. Include health teaching.	Preform head to toe with genitalia last

PHYSICAL ASSESSMENT OF THE NEWBORN

Usual Findings (Hockenberry & Wilson, 2009)

1. General measurements.

a. *Head circumference*—33-35 cm (13-14 inches); about 2-3 cm (1 inch) larger than chest circumference
b. *Chest circumference*—30.5-33 cm (12-13 inches)
c. *Crown-to-rump length*—31-35 cm (12.5-14 inches); approximately equal to head circumference
d. *Head-to-heel length*—48-53 cm (19-21 inches)
e. *Birth weight*—2700-4000 g (6-9 pounds)

2. Vital signs

a. *Temperature*
b. Axillary—36.5°-37° C (97.9°-98.6° F)
c. *Heart rate*
d. Apical—110-160 beats/min
e. *Respirations*
f. 30-60 breaths/min
g. *Blood pressure (BP)*
h. Oscillometric—65/41 mm Hg (average) in arm and calf

3. General appearance

a. *Posture*—Flexion of head and extremities, which rest on chest and abdomen

4. Skin

a. At birth, puffy, smooth
b. Second to third day, pink, flaky, dry
c. Vernix caseosa
d. Lanugo
e. Edema around eyes, face, legs, dorsa of hands, feet, and scrotum or labia
f. ***Acrocyanosis***—Cyanosis of hands and feet

5. Head

a. *Anterior fontanel*—Diamond-shaped, 2.5-4 cm (1-1.75 inches)

b. *Posterior fontanel*—Triangular, 0.5-1 cm (0.2-0.4 inch)
c. Fontanels should be flat and firm.
d. Widest part of fontanel measured from bone to bone, not suture to suture

6. Eyes

a. Lids usually edematous
b. Iris color—slate gray, dark blue, brown; absence of tears
c. Presence of red reflex
d. Corneal reflex in response to touch
e. Pupillary reflex in response to light
f. Blink reflex in response to light or touch
g. Rudimentary fixation on objects and ability to follow to midline

7. Ears

a. Position—Top of pinna on horizontal line with outer canthus of eye
b. Startle reflex elicited by a loud, sudden noise
c. Pinna flexible, cartilage present

8. Nose

a. Nasal patency
b. Nasal discharge—Thin, white mucus
c. Sneezing
d. Milia

9. Mouth and throat

a. Intact, high-arched palate
b. Uvula in midline
c. Frenulum of tongue
d. Frenulum of upper lip
e. Sucking reflex—Strong and coordinated
f. Rooting reflex
g. Gag reflex
h. Extrusion reflex
i. Absent or minimal salivation
j. Vigorous cry

10. Neck

a. Short, thick, usually surrounded by skin folds
b. Tonic neck reflex

11. Chest

a. Anteroposterior and lateral diameters equal
b. Slight sternal retractions evident during inspiration
c. Xiphoid process evident
d. Breast enlargement

12. Back and rectum

a. Spine intact; no openings, masses, or prominent curves
b. Trunk incurvation reflex
c. Anal reflex

d. Patent anal opening
e. Passage of meconium within 48 hours

13. Extremities

a. Ten fingers and ten toes
b. Full range of motion
c. Nail beds pink, with transient cyanosis immediately after birth
d. Creases on anterior two thirds of sole
e. Sole usually flat
f. Symmetry of extremities
g. Equal muscle tone bilaterally, especially resistance to opposing flexion
h. Equal bilateral brachial and femoral pulses

14. Neuromuscular system

a. Extremities usually maintain some degree of flexion.
b. Extension of an extremity followed by previous position of flexion
c. Head lag while sitting, but momentary ability to hold head erect
d. Able to turn head from side to side when prone
e. Able to hold head in horizontal line with back when held prone

Equipment of physical assessment



Thermometer (for measuring temperature)



Stethoscope (for measuring blood pressure and listening to heart, lung, and bowel sounds)



Sphygmomanometer (for measuring blood pressure)



Pocket ruler (to measure size or distance)



Tape measure (to measure circumference and length)



Scale (to measure weight and height)



Gloves (to wear if there is any possible exposure to blood or body fluids)

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Anthropometric measurements

Objectives

1- Defining the term of anthropometric measurements.
2- Identify measures to assess child growth and development status, included weight, height, head circumference, chest circumference.

3- Determine the measured process of anthropometric measurements in children according to the child age.

4- Assess the progression of weight and height, head and chest circumference, body max index.

Anthropometric measurements

Anthropometry: an essential parameter of **nutritional status**, is the measurement of height or length, weight, head circumference, chest circumference proportions, skinfold thickness, and arm circumference in young children. Height and head circumference reflect past nutrition, whereas weight, skinfold thickness, and arm circumference reflect present nutritional status, especially of protein and fat reserves.

Weight measured:

1- At birth, weight **2700** to **4000** kg (6–9 pounds).
2- First, balance the scale.
3- Remove the infant's diaper and clothing.
4- Depending on age, put the child is either supine or sitting.
5- Protect the infant from an accidental fall by placing a hand over the infant without direct contact.
6- At birth, weight is more variable than height and is, to a greater extent, a reflection of the intrauterine environment.
7- Normally, neonates lose about 10% of their birth weight by 3 to 4 days of age because of loss of extracellular fluid and meconium, as well as limited food intake, especially in breastfed infants.
8- In general, the birth weight doubles by 4 to 7 months of age, triples by the end of the first year. Quadruples by the age of 2 to 2.5 years.

Weight progression (1-12 month)

First 4 month = 750 gm /month
2nd 4 months = 500 gm /month
3rd 4 months = 250 gm/month
Progression in year: Weight (kg/years) = (age + 4) ×2.
Thus an average 2-year-old will weigh: (2 + 4) ×2 = 12kg.

Height or Length

1- The term length refers to measurements taken when children are supine
2- Length in infants are measured by lying in supine position on tray or board.

3- Hold the infant's head in midline and extend the hips and knees flat on a paper-covered table, then Points are marked at the top of the head and the heels of the feet

4- The term height (or stature) refers to the measurement taken when a child is standing upright.

Height:

Birth – 6 months: Monthly gain—2.5 cm (1 inch)

6-12 month: Monthly gain—1. 5 cm (0.5 inch)



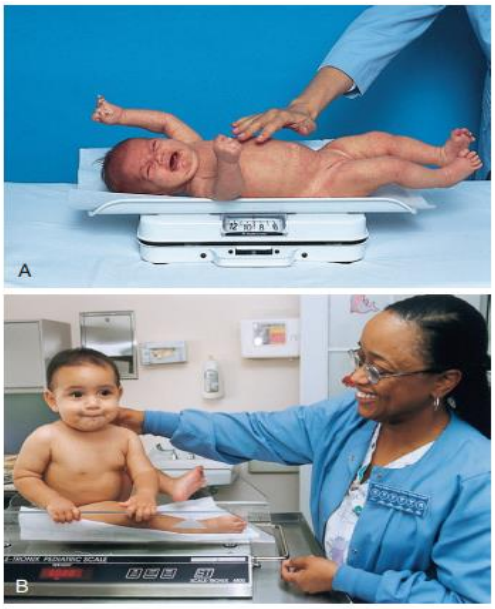
A. Measuring a newborn's length



B. Weighing the newborn



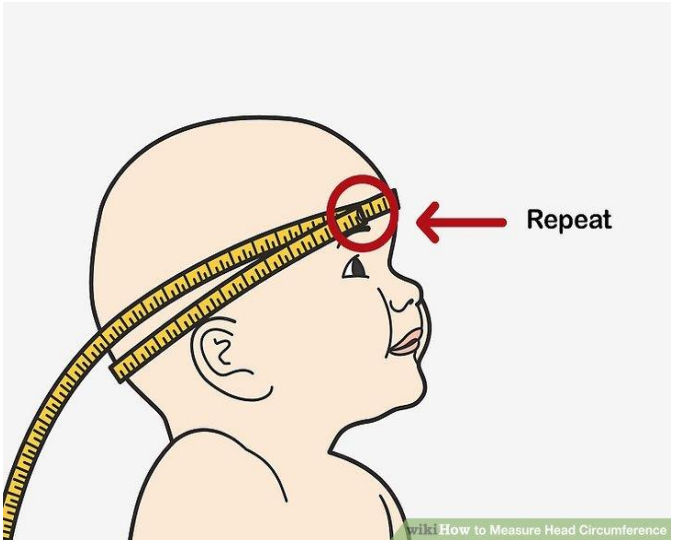
C. Measuring a child height.



D. Measurements (A. length, B. height)

Head circumference:

1- Head circumference is a critical indicator of brain growth in infants and toddlers.
2- Full-term infants, average head circumference is between 33 and 35.5 cm (13 and 14 inches).
3- Measure head circumference in children up to 36 months of age
4- The rate of increase in head circumference slows somewhat by the end of infancy, and head circumference is usually equal to chest circumference by 1 to 2 years of age.
5- The usual total increase in head circumference during the second year is 2.5 cm (1 inch). Then the rate of increase slows until at age 5 years, the increase is less than 1.25 cm (0.5 inch) per year.
6- The head's largest circumference is measured by placing the tape over the lower forehead, above the pinna of the ears, and over the occipital prominence.
7- Head circumference is usually about 2 to 3 cm (≈1 inch) greater than chest circumference.
8- Because of the molding of the head during delivery, these measurements may initially appear equal. However, if the head is significantly smaller than the chest, **microcephaly** or premature closure of the sutures (**craniosynostosis**) is a possibility. The head is more than 4 cm (1.6 inches) larger than the chest and this remains constant or increases over several days, then hydrocephalus must be considered.
9- During the first 6 months, head circumference increases approximately 1.5 cm (0.6 inch) per month, but the rate of growth declines to only 0.5 cm (0.2 inch) monthly during the second 6 months.
10- Closure of the cranial sutures occurs, with the posterior fontanel fusing by **6 to 8** weeks of age and the anterior fontanel closing by **12 to 18** months of age (average, 14 months).

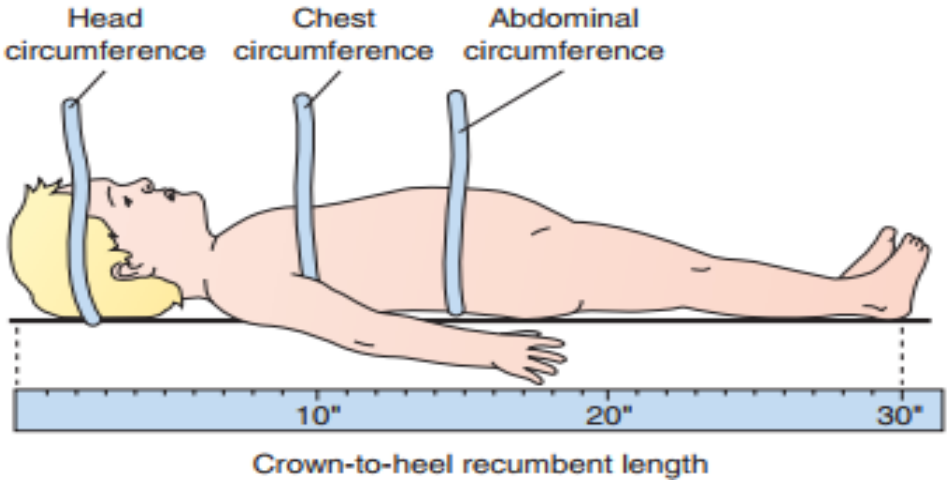


*H.C (Birth – 6 months): Monthly gain—1.5 cm

*H.C (6-12 month): Monthly gain—0. 25 cm

Chest circumference:

1- At birth, the chest circumference is about 30 to 33 cm or 12 to 13 inches. During year 2, the chest circumference continues to increase and eventually exceeds head circumference. the chest circumference exceeds the abdominal measurement after the second year, in addition to the growth of the lower extremities, makes the child appear taller and leaner.
2- Chest circumference continues to increase in size and exceeds head circumference during the toddler years.
3- The chest's shape also changes as the transverse, or lateral, diameter exceeds the anteroposterior diameter.
4- Toddlers retain a squat, "potbellied" appearance because of their less developed abdominal musculature and short legs.
5- The rate of physical growth slows and stabilizes during the preschool years.
6- Continues to increase in size and exceeds head circumference during the toddler years.



(Measurement of head, chest, abdominal circumference)

Body Mass Index

1- BMI is also a part of assessing growth and can be tracked over time.
2- BMI is the number calculated from the child's weight and height that correlates to the percentage of body fat. The higher the child's BMI, the higher the percentage of body fat.
3- (BMI [weight in kilograms divided by height in meters squared])

Parameters:

- Underweight: BMI of less than 18.5
- Normal weight: BMI of 18.5 to 24.9
- Overweight: BMI of 25 to 29.9
- Obesity: BMI of 30 or greater

BODY MASS INDEX FORMULA

BMI= Weight in kilograms÷ (Height in meters)^2

Physiologic measurements

Objectives

1- Defining the terms of physiologic measurements.
2- Identify key elements in evaluating the physical status of vital functions, include temperature, pulse, Respiration, and BP.
3- Demonstrate the temperature and pulse measurement sits for infants and children.
4- Recorded the type of thermometer.
5- Discuss the process of measuring the temperature, pulse, respiration, blood pressure in children according to the child age.

PHYSIOLOGIC MEASUREMENTS

Physiologic measurements: Its key elements in evaluating physical status of vital functions, include **temperature, pulse, Respiration, and BP.**

Temperature

- The core temperature most closely reflects the temperature of the blood flow through the carotid arteries to the hypothalamus.
- When a child's temperature is altered, receptors in the skin, spinal cord, and brain respond in an attempt to achieve **normothermia**, a normal temperature state.
- For neonates, a core body temperature between 36.5° and 37.6° C (97.7° and 99.7° F).
- One of the most important influences on the accuracy of temperature is improper temperature-taking technique.
- Because children generally dislike rectal temperature measurement, tympanic thermometers may be used for infants who are more than 3 months old, and oral thermometers may be used for children more than 4 to 5 years old.
- The axial route is preferred, but if a rectal temperature is required, carefully insert a well-lubricated thermometer no further than 1/2 to 1 inch to avoid rectal perforation.

Fahrenheit to Centigrade Temperature Conversion Formulas

°F = (°C × 9/5) + 32, or (°C × 1.8) + 32
°C = (°F - 32) × 5/9, or (°F - 32) × 0.55

Temperature measurement locations for infants and children: Included: oral, rectal, axillary, ear canal, tympanic membrane, temporal artery, or skin route.

Oral site:

- ✓ Place tip under tongue in right or left posterior sublingual pocket, not in front of tongue. Have child keep mouth closed without biting on thermometer.
- ✓ Pacifier thermometers measure intraoral or supralingual temperature and are available but lack support in the literature.
- ✓ Several factors affect mouth temperature: eating and mastication, hot or cold beverages, open-mouth breathing, ambient temperature.

Axillary site

- ✓ Place tip under arm in center of axilla and keep close to skin, not clothing. Hold child's arm firmly against side.
- ✓ Temperature may be affected by poor peripheral perfusion (results in lower value), clothing or swaddling, use of radiant warmer, or amount of brown fat in cold-stressed neonate (results in higher value).
- ✓ Advantage—Avoids intrusive procedure and eliminates risk of rectal perforation.



A. Axillary site B. Oral site

Ear Based (Aural) site

- ✓ Insert small infrared probe deeply into canal to allow sensor to obtain measurement. Size of probe (most are 8 mm) may influence accuracy of result. In young children, this may be a problem because of small diameter of canal. Proper placement of ear is controversial related to whether the pinna should be pulled in manner similar to that used during otoscopy.

Rectal site

- ✓ Place well-lubricated tip at maximum 2.5 cm (1 inch) into rectum for children and 1.5 cm (0.6 inch) for infants; securely hold thermometer close to anus.
- ✓ Child may be placed in side-lying, supine, or prone position (i.e., supine with knees flexed toward abdomen); cover penis because procedure may stimulate urination. A small child may be placed prone across parent's lap.

Temporal Artery site

- ✓ An infrared sensor probe scans across forehead, capturing heat from arterial blood flow.

✓ Temporal artery is only artery close enough to skin's surface to provide access for accurate temperature measurement.



c. Ear Based (Aural) site D. Temporal Artery site

Type of thermometer:

- ❖ mercury glass thermometers
- ❖ Digital Thermometer / Electronic Thermometer
- ❖ Infrared Thermometer
- ❖ Ear-Based Temperature Sensor

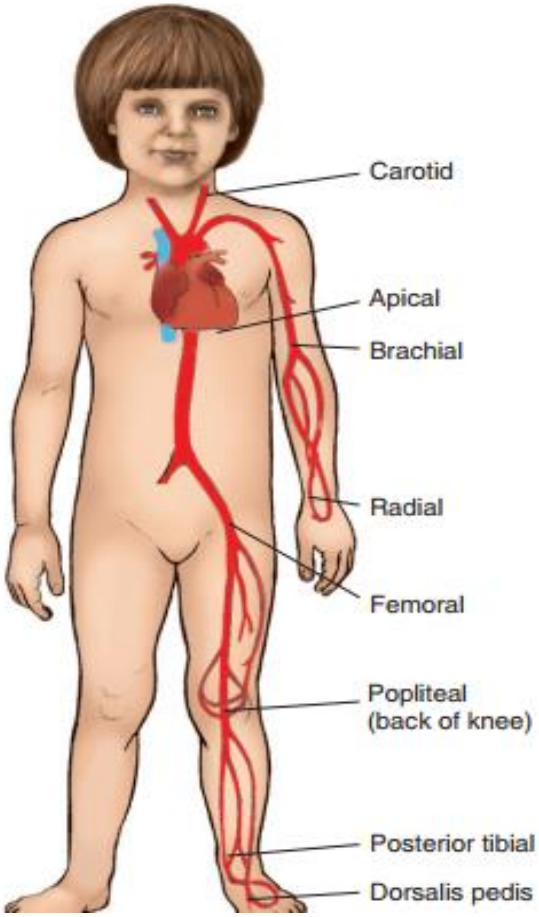
- Advantages of digital thermometers in neonatal care include relatively easy readability by parents and caretakers in the home,
- Infrared axillary and digital thermometers are used in many neonatal units because they give rapid readings and are easy to clean.
- In newborn temperature, the glass mercury thermometer is the gold standard against which other methods are compared.

Pulse

- ❖ A satisfactory pulse can be taken radially in children older than 2 years of age. However, in infants and young children, the **apical impulse** (heard through a stethoscope held to the chest at the apex of the heart) is more reliable**.**
- ❖ Count the pulse for 1 full minute in infants and young children because of possible irregularities in rhythm.
- ❖ Compare radial and femoral pulses at least once during infancy to detect the presence of circulatory impairment, such as coarctation of the aorta.
- ❖ At birth, the range of pulse 120 to 140 beats/min

- Auscultate the **apical pulse** by using the diaphragm of the stethoscope over the point of maximal intensity (PMI)**,** The PMI is located just left of the midclavicular line at the fourth intercostal space in children less than 7 years old and at the fifth intercostal space in older children.
- An apical pulse (listening at the heart apex through a stethoscope) is taken in children younger than 1 year because their radial (wrist) pulse is too faint to be palpated accurately.
- **Heart rate and rhythm**—Listen for fast heart rates (tachycardia), slow heart rates (bradycardia), and irregular rhythms

Locations of pulse



Respiration

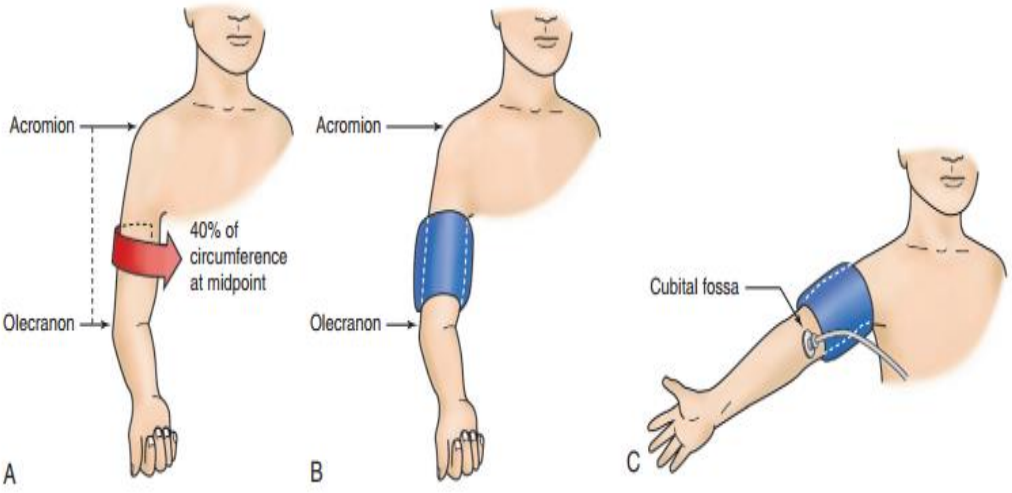
- At birth, respiratory rate average 30 to 60 breaths/min.
- Respirations also should be measured before an infant is disturbed because the respiratory rate increases with crying.

- Take this while a child is sitting in a parent's lap or lying quietly in a crib before lowering the side rail if possible
- Infants tend to breathe with their abdominal muscles; therefore, it is as accurate to take respirations by counting movements of the abdomen as it is to count chest movements.

Count respirations for 1 full minute.

Blood Pressure

- Blood pressure is included in the routine physical assessment of all children older than 3 years of age.
- Blood pressure is difficult to measure in infants because of mechanical problems.
- For healthy term infants, blood pressure average 70-65/45 mm Hg.
- Because measurement of blood pressure in a newborn is somewhat inaccurate, it is not routinely measured unless a cardiac anomaly is suspected, renal disorder such as, nephrotic syndrome and nephritis.



A. Determination of proper cuff size.

B. Cuff bladder width should be approximately 40% of circumference of arm.

C. Blood pressure should be measured with the cubital fossa at the heart level. The arm should be supported. The stethoscope bell is placed over the brachial artery pulse proximal and medial to the cubital fossa and below the bottom edge of the cuff.

Reflexes

Objectives

1- Defining the importance of reflexes assessment in children.
2- Demonstrate how to assess reflexes in infants.
3- Discuss the type of reflexes(sucking, rooting, Babinski, startle and Moro reflexes …etc.).
4- Explain the child's response to each reflex according to the offered type.

Reflexes

- Testing reflexes is an important part of the neurologic examination.
- Reflexes should be tested at the end of the examination because they may disturb the infant and interfere with auscultation.
- Reflexes are present at birth and disappear by 9 months of age.
- **Primitive reflexes** originate in the central nervous system and are exhibited in normal infants.

1. **Rooting reflex:** The infant turns his head toward the mother's breast and begins to suck when his lower lip or cheek is stroked.
2. **Sucking reflex:** The infant moves his lips, mouth, and tongue and shows interest to feed. Present at birth; disappears at 10–12 months.
3. **Startle reflexes:** Make a loud sound near the neonate.
4. **The tonic neck reflex** (or fencing posture): The infant turns his head to one side, and the arm on that side straightens while the opposite arm bends in a fencing position.
5. **The Moro reflex:** Jar the crib or hold the baby in a semi-sitting position and let the head slightly drop back.
6. **The grasping reflex**: An object is placed in the infant's hand, and the infant's fingers close to grasp the object.
7. **The Babinski reflex:** The sole of the foot is stroked, and the infant responds by turning the foot inward while his toes flare outward or flexion.
8. **Dancing (or stepping) reflex:** The soles of the infant's feet touch a flat surface, and he attempts to walk by placing one foot in front of the other.
9. **Palmar grasp:** The examiner places a finger in the palm of the neonate's hand, he neonate grasps the finger tightly; if the neonate grasps the examiner's finger with both hands, the neonate can be pulled to a sitting position.
10. **Plantar grasp:** Place a thumb firmly against the ball of the infant's foot, Toes flex tightly down in a grasping motion.
11. **Protective reflexes:** Gag, cough, and sneeze reflexes also are present in newborns to maintain a clear airway.



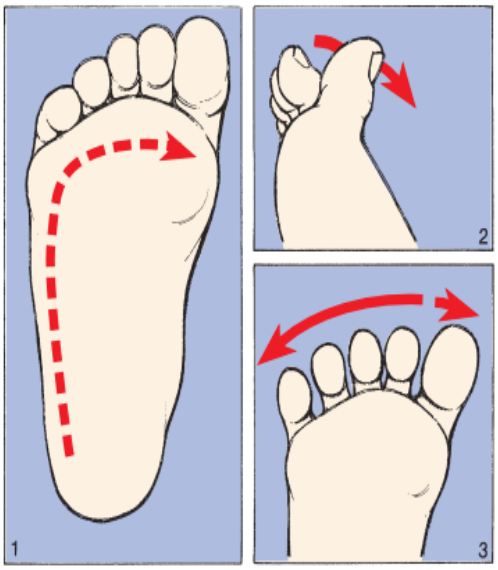
A, Moro reflex. **B,** Tonic neck reflex. **C,** Dance reflex. **D,** Crawl reflex.



E. Moro reflex F. Grasp reflex



G. Babinski reflex



H. Direction (**1.** Direction of stroke. **2,** Dorsiflexion of big toe. **3,** fanning of toes)

References:

1- Hockenberry, M. J., Wilson, D.(2015). Wong's nursing care of infants and children, 10th edition. *Canada: Elsevier*, P.P 113-124.

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3- Pillitteri, A. (2010). Maternal and child health nursing: care of the childbearing and childrearing family, 6th edition. *Philadelphia: Lippincott Williams & Wilkins*, p p 450-453.

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Drug Administration

Objectives:

By the end of this chapter, the student nurse will be able to:

1- Define the meaning of medications.

2- Calculate a safe drug dosage needed for the individual child.

3- Identify the factors that influence medications action.

4- List the essential principles for the administration of medications.

5- Enumerate the different routes of medications administration.

6- Explain preliminary information essential for the administration of the different routes:

7- Discuss the steps of the different routes of administration of medications.

8- Describe the role of the nurse in each route.

Introduction:

Because a child responds to drugs more rapidly and unpredictably than an adult does, pediatric drug administration requires special care. Such factors as age, weight, body surface area, and drug form and route may dramatically affect a child's response to a drug. For example, because of his thin epithelium, a neonate or an infant absorbs topical medications much faster than an older child does.

Usual drug administration techniques may need adjustment to account for the child's age, size, and developmental level. A tablet for young child, for example, may be crushed and mixed with a liquid for oral administration. In addition, the injection site and needle size will vary depending on the child's age and physical development.

Usual drug administration techniques may need adjustment to account for the child's age, size, and developmental level. A tablet for a young child, for example, may be crushed and mixed with a liquid for oral administration. In addition, the injection site and needle size will vary depending on the child's age and physical development.

General steps for administer medication

- ☒ Assess the child's condition to determine the need for the medication and the effectiveness of previous therapy.
- ☒ Carefully observe the child for a rash, pruritus, cough, or other signs of an adverse reaction to a previously administered drug.
- ☒ Identify the child by comparing the name on his wristband with that on the medication card. If the child can talk and respond, ask him his name.
- ☒ Explain the procedure to the child and his parents. Use terms the child can understand.
- ☒ Provide privacy, especially for an older child.

Administering Oral Medications

ESSENTIAL STEPS for oral medication

➢ Check the physician's order for the prescribed drug and dosage.

➢ Compare the order with the drug label, check the drug expiration date, and review the child's chart for any drug allergies.

➢ Carefully calculate the dosage. Double-check dosages with another nurse for potentially hazardous or lethal drugs, such as insulin, heparin, digoxin, epinephrine, and opioids. Check your

facility's policy to learn which drugs must be calculated and checked by two nurses.

➢ Confirm the child's identity using two patient identifiers according to facility policy.

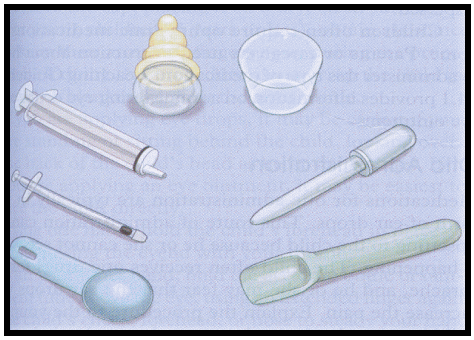
Purpose

To safely administer oral medications

Equipment

For oral medications

- ❖ Prescribed medication
- ❖ Plastic disposable syringe, plastic medicine dropper, or spoon
- ❖ Medication cup
- ❖ Water, syrup, or jelly (for tablets)
- ❖ Optional: fruit juice.



Steps

1. Verify medication orders and assemble supplies.
2. Place the infant's legs between your knees to maintain control of them.
3. Place one of the infant's arms behind your back and restrain the other arm by "hugging" the child (this helps to prevent the child from tipping the medication cup).

4. Give the medication slowly by cup or syringe to allow the child time to swallow. An older child may be given the choice to self-administer the medication by cup or syringe.
5. Provide "chasers" of water, fruit juice, or frozen ice-pops to help lessen the residual taste of the medicine. The patient's age and diet prescription are considered when choosing a chaser.
6. Remove gloves and perform hand hygiene.
7. Document in the medication administration record (MAR) the medication, route, time, and dose given

- **Measure a drug ordered in teaspoons in milliliters; the established standard is 5 ml/tsp.**

Notices

- The administration of medication by mouth is preferred in children but is not always possible because of vomiting, malabsorption, or refusal.

- Only scored tablets should be divided. Extended-release tablets should not be chewed or crushed. Gel tablets should not be cut or dissolved. Suspensions must be fully shaken before use.

- Medications should not be diluted in formula or water, because medications should be given in the smallest amount possible to assure the complete dose is consumed.

- The use of important sources of nutrients, foods, or liquids (e.g., orange juice) for disguising the taste of the medication is discouraged because the child may develop a distaste for these foods.

- Medication is administered slowly, especially if the child is crying. The child's head and shoulders are elevated to prevent aspiration. Toddlers may attempt to push away the medicine cup. In anticipation of this response, the nurse holds the child in a "hug" position in the nurse's lap in a semi sitting position

- If a nasogastric tube is in place, the nurse tests for proper placement of the tube before pouring medication into the funnel. A small amount of water is administered afterward to flush (cleanse) the tube.

- An oral syringe is an excellent device for measuring small quantities. It is easily transported, and medication can be provided directly from the syringe. The syringe is placed midway back at the side of the mouth.



FIG. 22.9 The Medibottle is attached to the syringe so that the infant can suck on the nipple to consume the medication. The nurse controls the flow with gentle pressure on the syringe barrel. (Courtesy The Medicine Bottle Co.)



The syringe or cup method of administering oral medications to infants or children. Note that one of the infant's arms is behind the nurse's back while the nurse restrains the other. The legs of the infant can be held firmly between the knees of the nurse for better control during medication administration.

Giving oral medication to an infant

- Use a plastic syringe without a needle or a drug-specific medicine dropper to measure the dose. If the medication comes in tablet form, first crush the tablet (if appropriate)

and mix it with water or syrup. Then draw the mixture into the syringe or dropper.

- Pick up the infant, raising his head and shoulders or turning his head to one side

R:to prevent aspiration.

- Hold the infant close to your body

R: To help restrain him.

- Using your thumb, press down on the infant's chin

R: to open his mouth.

- Slide the syringe or medicine dropper into the infant's mouth alongside his tongue. Release the medication slowly

R:to let the infant swallow and to prevent choking. If appropriate, allow him to suck on the syringe as you expel the medication.

- If not contraindicated, give fruit juice after giving medication.
- Then place a particularly small or inactive infant on his side or back

R:to decrease the risk of sudden infant death syndrome, as recommended by the American Academy of Pediatrics.

- Allow an active infant to assume a position that's comfortable for him; avoid forcing him into a side lying position

R:to prevent agitation.

Giving oral medication to a toddler

- Use a plastic, disposable syringe or dropper to measure liquid medication. Then transfer the fluid to a medication cup.
- Elevate the toddler's head and shoulders

R: to prevent aspiration.

- If possible, ask him to help hold the cup

R:to enlist his cooperation. Otherwise, hold the cup to the toddler's lips, or use a syringe or a spoon to administer the liquid. Make sure the toddler ingests all of the medication.

- If the medication is in tablet form, first crush the tablet, if appropriate, and mix it with water, syrup, or jelly. Use a spoon, syringe, or dropper to administer the medication.

Giving oral medication to an older child

- If possible, let the child choose the liquid medication mixer and a beverage to drink after taking the medication.
- If appropriate, allow him to choose where he'll take the medication, for example, sitting in bed or sitting on a parent's lap.
- If the medication comes in tablet or capsule form, and if the child is old enough (between ages 4 and 6), teach him how to swallow solid medication. (If he knows how to do this, review the procedure with him for safety's sake.) Tell him to place the pill on the back of his tongue and to swallow it immediately by drinking water or juice. Focus most of your explanation on the water or juice.

R:to draw the child's attention away from the pill. Make sure the child drinks enough water or juice to keep the pill from lodging in his esophagus. Afterward, look inside the child's mouth

R: to confirm that he swallowed the pill.

- If the child can't swallow the pill whole, crush it (if appropriate) and mix it with water, syrup, or jelly. Alternatively, after checking with the child's physician, order the medication in liquid form.

Guidelines for administering oral medication

- Do not say, "*Can* you drink this for me?" If an adult seems unsure whether children can do a task, children can develop

grave doubts themselves or they may just not want to and say "No."

- Do not say, "*Will* you drink this for me?" This leaves a child the opportunity to say no and creates the awkward position of having to admit the child really does not have a choice in the matter; the child must take the medicine.
- State firmly, "It's time for you to drink your medicine now." Give the child a secondary choice to allow a sense of control: "It's time to drink your medicine now; do you want milk or water to swallow after it?" is a suitable choice, assuming both milk and water are compatible with the medication.
- Never refer to medicine as candy. If they think it is candy, children may help themselves to fatal amounts of medicine when everyone's back is turned.
- Do not use reward systems to persuade children to take medicine. Bribing may work for one dose, but when a second dose is due, the child will ask for a bigger bribe; for a third dose, an even bigger one. At some point (generally reached quickly), it is impossible to supply such large bribes, and therefore, it is impossible to gain the child's cooperation.
- Do not threaten. Statements such as "Take this quickly or I'll ask the doctor to make it into a shot" are not appropriate or therapeutic.

- Children expect honesty from adults; therefore, do not lie about the taste of medicine. If in doubt about the taste, taste it (with the obvious exception of drugs such as digoxin). Most children's medicines are artificially flavored with raspberry, orange, or cherry syrup, so they do not taste bad.
- If a medicine tastes bitter, mix it with a spoonful of strained applesauce or a teaspoonful of flavored syrup. Do not mix medicine with a full jar of baby food or put it into a bottle of milk or juice because the child will then have to eat the entire jar of food or drink the entire bottle to get all of the medicine. As a rule, encourage children to take liquid medicine straight and then follow it with a pleasant-tasting drink to take away any bitter taste. You could offer the choice of taking it in syringe or a medicine cup to provide some control.
- If a medicine is supplied in tablet form, you could crush or dissolve it in with syrup or applesauce for a better taste if appropriate. Be certain before removing the particles from a capsule that the medicine will work properly when not in capsule form: Some are encapsulated to keep them from dissolving in the stomach and to bring them into the intestine, where they have a therapeutic effect. The same precaution must be followed when giving enteric coated tablets.
- Never leave medicine by a child's bed or chair for the child to take "in a minute" or "after your shower." The child may become involved with another activity "in a minute"

and will not take it or a smaller child could find the medicine appealing and swallow it.

Administering parenteral medications

Nose Drops, Ear Drops, and Eye Drops

Except for a few differences, the principles for administering nose drops, ear drops, and eye drops to children are essentially the same as for adults. Infants and small children may need to be restrained in a mummy restraint

NURSING TIP

To reduce unpleasant sensations when administering medications:

• **Eye**—Apply finger pressure to the lacrimal punctum at the inner aspect of the eyelid for 1 minute to prevent drainage of medication to the nasopharynx and the unpleasant "tasting" of the drug.

• **Ear**—Allow medications stored in the refrigerator to warm to room temperature before instillation.

• **Nose**—Position the child with the head hyperextended to prevent strangling sensations caused by medication trickling into the throat rather than up into the nasal passages.

Nose drops

The procedure for administering nose drops to a small child

Purpose

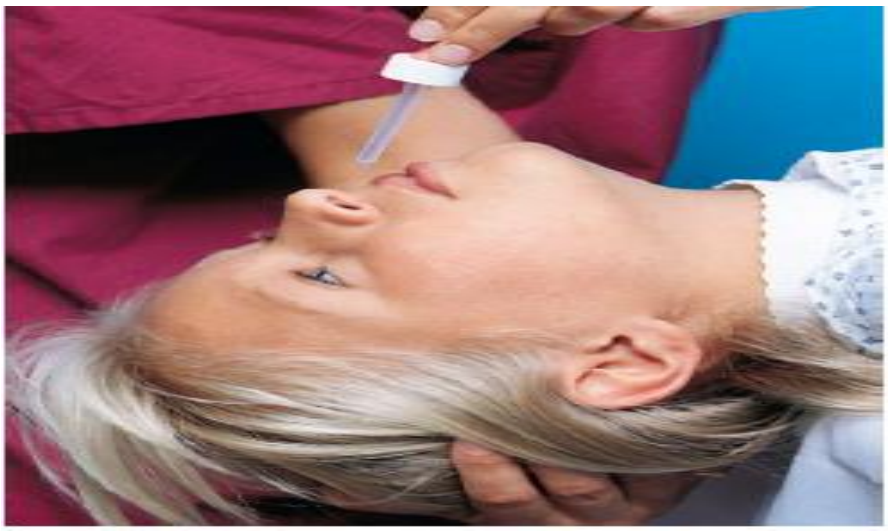
To administer medication by nasal route

Steps

1. Immobilize the infant with a mummy restraint.
2. Wipe excess mucus from nose with a tissue.
3. Place the infant on his or her back, with the head over the side of the mattress or the neck extended over a pillow.
4. Encircle the infant’s cheeks and chin with the left arm and hand to steady.
5. Instill drops with the right hand.
6. Keep infant in this position for 30 seconds to 1 minute to allow the drops to reach the proper area.
7. Remove restraints. Make the infant comfortable.
8. Remove gloves and perform hand hygiene.
9. In the patient’s chart, record: time, name of nose drops, strength, and number of drops instilled, how the patient tolerated the procedure, and untoward reactions.

Ear drops

The health care provider may



prescribe a drug to be instilled into the ear to relieve pain. If the drops were refrigerated, they are allowed to warm to room temperature before administration.

Purpose

To administer medications into the ear canal

Steps

1. Place child in supine position with unaffected ear down.
2. Instill the ordered number of drops:

a. In children less than 3 years of age, pull the pinna (earlobe) of the affected ear down and back to straighten the canal.



b. In older children, pull the upper pinna (auricle) up and back to straighten the canal.

3. Gently massage the area in front of the ear to facilitate entry of the drops.
4. Keep the patient in a supine position for a few minutes to permit the fluid to be absorbed.
5. Remove gloves and perform hand hygiene.
6. Document the time, name of drug, number of drops administered, area (right or left ear), untoward reactions, and whether or not the patient obtained relief.

Eye drops and creams

Purpose

To protect against ophthalmia neonatorum and chlamydia infections (all newborns have this prophylactic treatment before leaving the delivery room)

Steps:

- Ophthalmic medication is administered to a child in the same manner as for the adult
- The child is informed of the need for the medication.
- The patient is identified, and the orders and the label on the bottle are checked for correct medication and concentration.



- The nurse ascertains which eye requires treatment.
- Hand hygiene is performed before and after the procedure, and clean gloves are worn. With the thumb and index finger, gentle pressure is applied in opposite directions to open the eye.
- The older child is instructed to "look up." The nurse supports his or her hand on the patient's forehead, and instills the medication into the center of the lower lid (conjunctival sac)
- The child is instructed to close the eye but not to squeeze it, because this could expel some of the solution.

Ointment

- ❖ Ointment is applied to the same conjunctival sac as eye drops.
- ❖ The mummy restraint is applied when only one nurse is available and the patient is an infant.
- ❖ Occasionally children refuse to open their eyes.
- ❖ The nurse must use ingenuity to coax reluctant children. It may help to involve the parents.

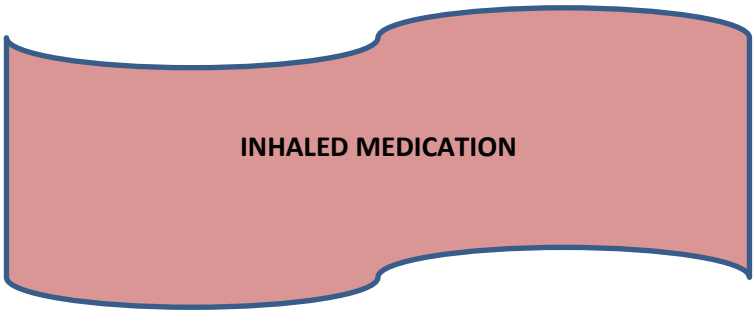
Drug Alert

If both eye ointment and drops are ordered, give drops first, wait 3 minutes, and then apply the ointment to allow each drug to work. When possible, administer eye ointments before bedtime or naptime because the child's vision will be blurred temporarily.

NURSING TIP

To administer oral, nasal, or optic medication when only one person is available to hold the child, use the following procedure:

- ➢ Place child supine on a flat surface (e.g., bed, couch, floor).
- ➢ Sit facing child so child's head is between operator's thighs and child's arms are under operator's legs.
- ➢ Place lower legs over child's legs to restrain lower body, if necessary.
- ➢ To administer oral medication, place a small pillow under child's head to reduce risk of aspiration.
- ➢ To administer nasal medication, place a small pillow under child's shoulders to aid flow of liquid through nasal passages.



Introduction

Administering inhaled medication to children through a nebulizer has the advantage of delivering a medication directly to the needed site of action. Advantages of administering inhaled medication with appropriate technique show an improvement in the onset of action, and a decrease in the dose needed.

Equipment

A pressurized canister
A plastic holder
A cap.
Often a spacer or special tube that attaches to the inhaler is used to increase the amount of medication delivered to the lungs per spray. A spacer has either a mouthpiece or a mask for ease of administration and is specific to the child's developmental need.

PREPARATION

Prepare to give inhaled medication by removing the cap from the inhaler and spacer as needed. Prime the inhaler by squirting a puff of medication into the air if the canister is new or has not been

used in the last 48 hours. If a spacer is used, insert the inhaler's mouthpiece into the rubber end of the spacer. Shake the inhaler immediately before each puff.

Procedure

- Placing the mouthpiece fully into the mouth between the teeth
- Hold the inhaler in the upright position and close the lips around the mouthpiece
- Press down on the canister to dispense the medication into the spacer
- Instruct the patient to breathe deeply for 3–5 seconds through the mouth, and then hold the breath for 10 seconds.
- Remove the spacer from the mouth and breathe out slowly.
- Patient should rinse the mouth out with water after each use of inhaled steroids to help prevent oral infections, and the spacer should routinely be washed with warm soapy water and air dried



- Teach families to consistently use proper technique with their children
- Often young children resist the mask when it is first introduced; however, with consistent practice eventually they will cooperate when they realize the parent is serious about giving their medications effectively.



Administering Intramuscular Injections

Purpose:

To place medication into the muscle without injury.

Equipment:

- ✓ Prescribed medication
- ✓ Appropriately sized syringe and needle
- ✓ Alcohol pads or povidone-iodine solution
- ✓ Gloves
- ✓ Gauze pads
- ✓ Cold compresses
- ✓ Adhesive bandage.

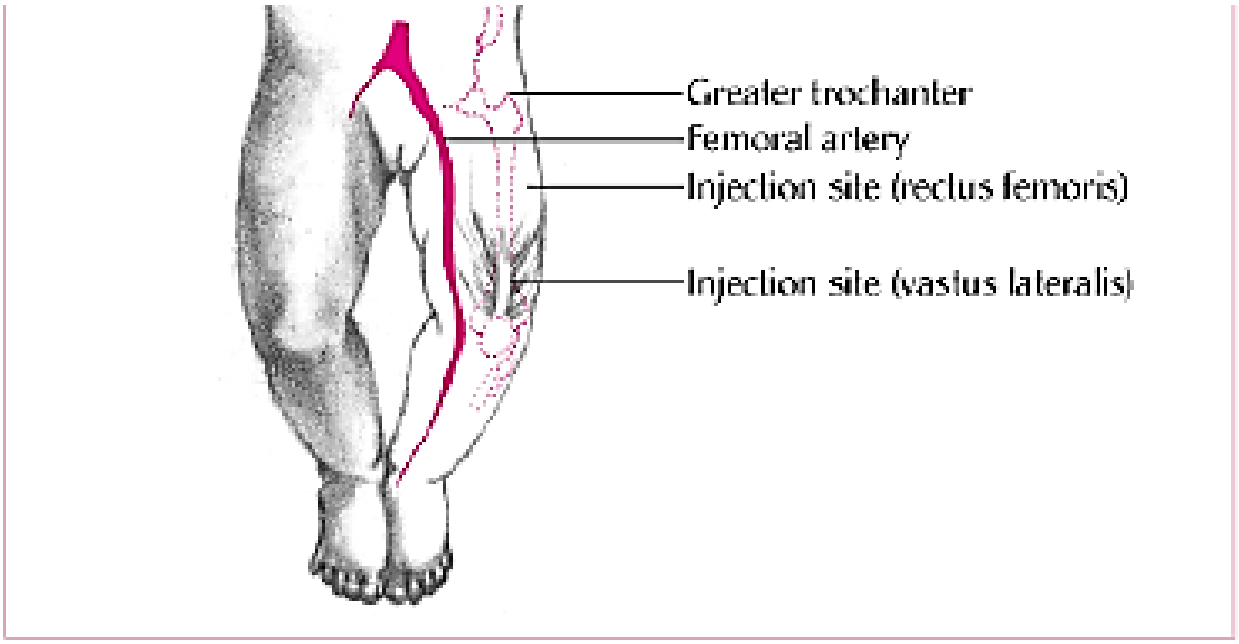
I.M. INJECTION SITES IN CHILDREN

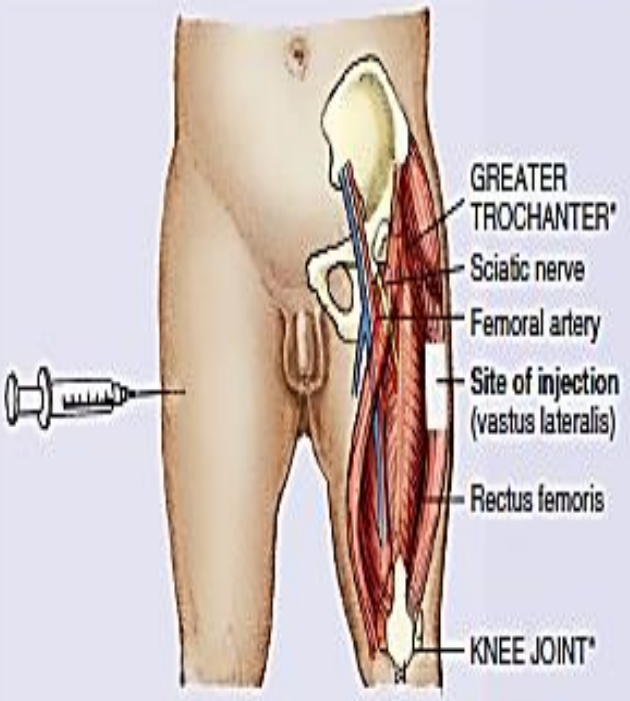
When selecting the best site for a child's I.M. injection, consider:

- the child's age, weight, and muscular development
- the amount of subcutaneous fat over the injection site
- the type of drug you're administering
- the drug's absorption rate.

Vastus lateralis and rectus femoris

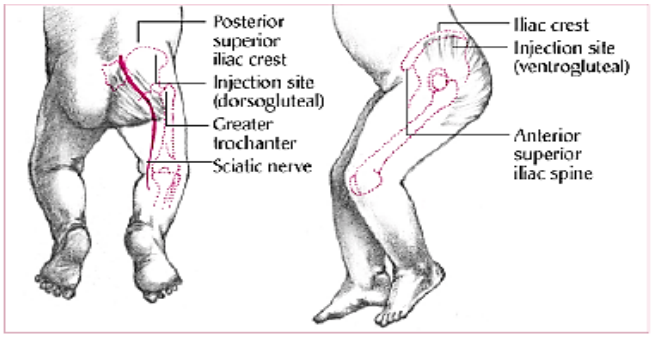
For a child under age 3, you'll typically use the vastus lateralis or rectus femoris muscle for an I.M. injection. Constituting the largest muscle mass in this age-group, the vastus lateralis and rectus femoris have fewer major blood vessels and nerves.



SITE	DISCUSSION
VASTUS LATERALIS	
	**Location*** Palpate to find greater trochanter and knee joints; divide vertical distance between these two landmarks into thirds; inject into middle third. **Needle Insertion and Size** Insert needle perpendicular to knee in infants and young children or perpendicular to thigh or slightly angled toward anterior thigh. 22-25 gauge (⅝-1 inch) **Advantages** Large, well-developed muscle that can tolerate larger quantities of fluid (0.5 ml [infant] to 2.0 ml [child]) Easily accessible if child is supine, side lying, or sitting **Disadvantages** Thrombosis of femoral artery from injection in midthigh area Sciatic nerve damage from long needle injected posteriorly and medially into small extremity More painful than deltoid or gluteal sites

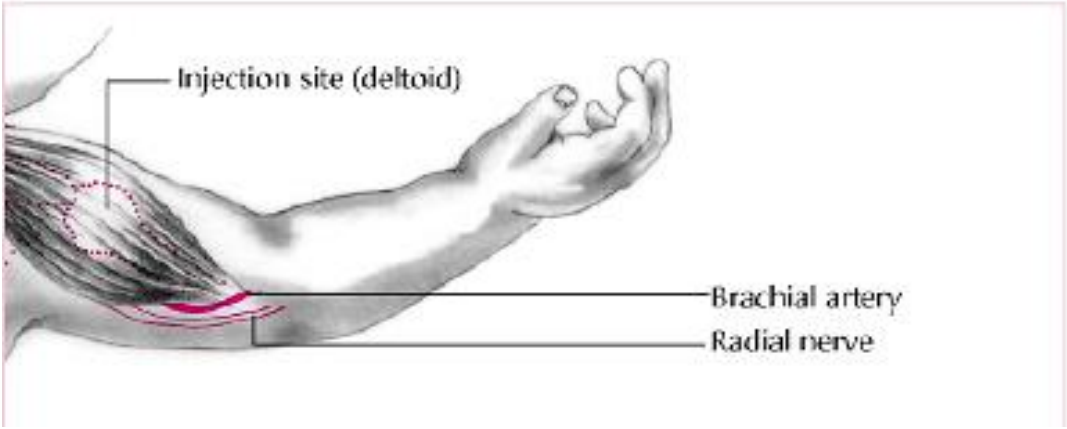
Ventrogluteal and dorsogluteal

For a child who can walk and is over age 3, use the ventrogluteal and dorsogluteal muscles. Like the vastus lateralis, the ventrogluteal site is relatively free from major blood vessels and nerves. Before you select either site, make sure that the child has been walking for at least 1 year to ensure sufficient muscle development.



Deltoid

For a child older than 18 months who needs rapid medication results, consider using the deltoid muscle for the injection. Because blood flows faster in the deltoid muscle than in other muscles, drug absorption should be faster. However, be careful when using this site because the deltoid doesn't develop fully until adolescence. In a younger child, it's small and close to the radial nerve, which may be injured during needle insertion.



SITE	DISCUSSION
VENTROGLUTEAL	
 Ventrogluteal site of injection	**Location*** Palpate to locate greater trochanter, anterior superior iliac tubercle (found by flexing thigh at hip and measuring up to 1-2 cm [0.4-0.8 inch] above crease formed in groin), and posterior iliac crest; place palm of hand over greater trochanter, index finger over anterior superior iliac tubercle, and middle finger along crest of ilium posteriorly as far as possible; inject into center of V formed by fingers. **Needle Insertion and Size** Insert needle perpendicular to site but angled slightly toward iliac crest. 22-25 gauge (⅝-1 inch) **Advantages** Free of important nerves and vascular structures Easily identified by prominent bony landmarks Thinner layer of subcutaneous tissue than in dorsogluteal site, thus less chance of depositing drug subcutaneously rather than intramuscularly Can accommodate larger quantities of fluid (0.5 ml [infant] to 2.0 ml [child]) Easily accessible if child is supine, prone, or side lying Less painful than vastus lateralis **Disadvantages** Health professionals' unfamiliarity with site
DELTOID	
	**Location*** Locate acromion process; inject only into upper third of muscle that begins about two finger breadths below acromion. **Needle Insertion and Size** Insert needle perpendicular to site but angled slightly toward shoulder. 22-25 gauge (⅝-1 inch) **Advantages** Faster absorption rates than gluteal sites Easily accessible with minimal removal of clothing Less pain and fewer local side effects from vaccines compared with vastus lateralis **Disadvantages** Small muscle mass; only limited amounts of drug can be injected (0.5-1.0 ml) Small margins of safety with possible damage to radial nerve and axillary nerve (not shown; lies under deltoid at head of humerus)

*Locations are indicated by asterisks on illustrations.

Procedure steps

1.Wash the infant's thigh if the bath has not yet been given. Blood from the mother is present on the infant's skin.

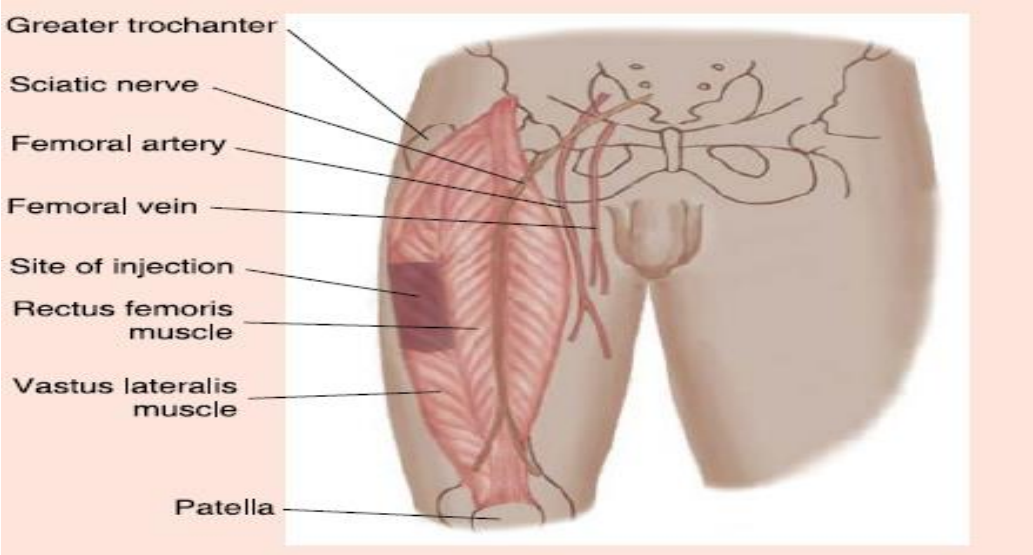
R: Washing prevents that blood from being carried into the infant's tissues during the needle insertion.

2. Prepare medication for injection. Use a 1mL syringe and in, 25gauge needle. If the medication is in a glass ampule, use a filter needle to draw it up. Remove the filter needle, and place a 25gauge needle on the syringe to give the injection. Use of a filter needle prevents particles of glass from being drawn into the syringe. A small need/e reaches the newborn's muscle but avoids the bone.

3. Put on gloves.

R: Gloves protect the nurse from contamination by blood.

4. Locate the correct site. The best site for newborn intramuscular injections is the vastus lateralis muscle. Divide the area between the greater trochanter of the femur and the knee into thirds. Give the injection in the middle third of the muscle, lateral to the midline of the anterior thigh. The large vastus lateralis is located away from the sciatic nerve and the femoral artery and vein. The rectus femoris muscle is nearer to these structures and poses more of a danger. (Note: The gluteal muscles are not used until a child has been walking for at least a year. These muscles are poorly developed and dangerous/y near the sciatic nerve.)



5. Cleanse the area with an alcohol wipe.

R: Cleansing removes organisms and prevents infection.

6. Stabilize the leg firmly while grasping the thigh between the thumb and fingers.

R: Holding the leg prevents sudden movement by the infant and possible injury

7. Insert the needle at a 90—degree angle. Inject the medication.

R: This places the medication into the muscle rather than into the subcutaneous tissue.



8. Withdraw the needle and apply gentle pressure to the site with small gauze pad. **R:** Massaging promotes absorption of the medication.

9. Discard the needle and syringe in the appropriate place and comfort the infant. **R:** The needle and syringe should be placed in the proper container to prevent injury or contamination to others. Infants should be comforted after painful procedures.

9- Choose an injection site that's appropriate for the child's age and muscle mass.

10- Position the patient appropriately for the site chosen, and locate key landmarks, for example, the posterior superior iliac spine and the greater trochanter. Have someone help you restrain an infant; seek an older child's cooperation before enlisting assistance.

11- Put on gloves. Clean the injection site with an alcohol or povidone-iodine pad. Wipe outward from the center with a spiral motion to avoid contaminating the clean area.

12- Grasp the tissue surrounding the site between your index finger and thumb to immobilize the site and to create a muscle mass for the injection.

13- Insert the needle quickly, using a darting motion. If you're using the ventrogluteal site, insert the needle at a 45-degree angle toward the knee.

14- Aspirate the plunger to ensure that the needle isn't in a blood vessel. If no blood appears, inject the medication slowly so that the muscle can distend to accommodate the volume.

15- Withdraw the needle and gently massage the area with a gauze pad to stimulate circulation and enhance absorption.

16-Provide comfort and praise.

Giving a subcutaneous injection

Procedure:

- Select from these possible sites: the middle third of the upper outer arm, the middle third of the upper outer thigh, or the abdomen. You may apply a cold compress to the injection site to minimize pain.
- Put on gloves, and prepare the injection site with alcohol or povidone-iodine solution according to the patient's needs and hospital policy.
- Pinch the tissue surrounding the site between your index finger and thumb to ensure injection into the subcutaneous tissue.
- Holding the needle at a 45- to 90-degree angle, quickly insert it into the tissue.
- Release your grasp on the tissue, and slowly inject the medication.
- Remove the needle quickly to decrease discomfort.
- Unless contraindicated, gently massage the area to facilitate the drug's absorption.

Giving an intradermal injection

Procedure:

✓ Put on gloves and pull the patient's skin taut (the site of choice is the inner aspect of the forearm).

✓ Insert the needle, bevel up, at a 10- to 15-degree angle just beneath the outer skin layer.

✓ Slowly inject the medication, and watch for a bleb to appear. Quickly remove the needle, being careful to maintain the injection angle. If appropriate—for example, if the injection is related to allergy testing—draw a circle around the bleb, and avoid massaging the area to avoid interfering with test results.

Special considerations when drug administration

❖ Don't hesitate to consult the parents for tips on successfully giving medication to their child. If possible, have a parent administer a prescribed oral drug while you supervise. However, avoid asking a parent to help with injections because the child may perceive the parent as a cause of pain.

❖ Aim for a trusting relationship with the child and his parents so that you can offer support and promote cooperation even when a medication causes discomfort. If the child will receive one injection, allow him to choose from the appropriate sites. However, if he'll receive numerous injections, remember that site rotation must follow a set pattern. Allow the child to play with a medication cup or syringe and to pretend to give medication to a doll.

❖ When giving medication to an older child, be honest. Reassure him that distaste or discomfort will be brief.

Emphasize that he must remain still to promote safety and minimize discomfort. Explain to the child and his parents that an assistant will help the child remain still if necessary. Keep your explanations brief and simple.

❖ To divert the child's attention, have him start counting just before the injection, and challenge him to try to reach 10 before you finish the injection. If the child cries, don't scold him or allow the parents to scold him. Have one of the parents hold a younger child and praise him for allowing you to give him the injection. You can also apply an adhesive bandage to the injection site as a form of reward or badge.

❖ If the prescribed medication comes only in tablet form, consult the pharmacist (or an appropriate drug reference book) to make sure crushing the tablet won't invalidate its effectiveness. Avoid adding medication to a large amount of liquid such as the child's milk or formula because the child may not drink the entire amount, resulting in an inaccurate dose of medication.

❖ Because infants and toddlers can't tell you what effects they're experiencing from a medication, you must be alert for signs of an adverse reaction. Compile a list of appropriate emergency drugs, calculating the dosages to the patient's weight. Post the list near the patient's bed for reference in an emergency.

❖ If you have any doubt about proper medication dosage, always consult the physician who ordered the drug. Double-check information in a reliable drug reference.

Administration of safe medication:

To administer medications safely, the following actions are necessary:

1- Accurately interpret the doctor's order.

2- Accurately calculate the amount of drug to be given for the prescribed dose.

3- Using the five rights for drug administration such as:

Right medication, Right dose, Right patient, Right time and Right route of administration.

4- Document medication administration according to agency policy.

Six right for administer medication

1-Right Medication

2-Right Patient

3-Right Dosage

To calculate a fractional drug dose use the formula:

$$\frac{\text{Strength Desired (D)}}{\text{Strength You Have (H)}} \times \frac{\text{Quantity Desired (QD)}}{\text{Quantity You Have (QH)}} = \text{Answer}$$

For example, you have a prescription for 90 mg acetaminophen. Acetaminophen is supplied as 125 mg drug in 5 ml liquid. Using the formula:

$$\frac{\text{90 mg (D)}}{\text{125 mg (H)}} \times \frac{\text{QD (what you are asking)}}{\text{5 ml (QH)}} = ?$$

Cross multiply: 125 QD = 450 (5 × 90)

Dose ordered = 90 mg divided by what you have on hand (how the medication is

available) 125 mg times quantity (5 ml)

$$\frac{\text{90 mg}}{\text{125 mg}} \times \text{5 ml} = \text{3.6 ml}$$

4- Right Route

5- Right Time

6-Documentation

Calculating Safe Dosages for Children

Calculating Safe Dosages for Children

STEPS

Dosage based on weight in kilograms

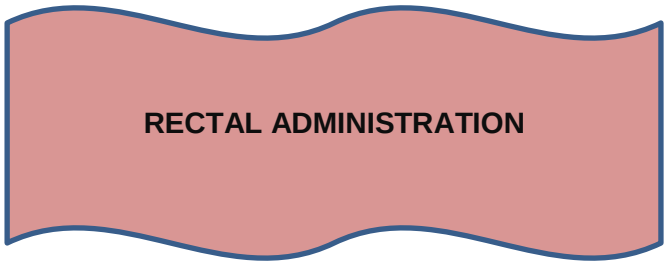
- If necessary, convert child's weight in pounds to kilograms. Child weighs 22 pounds.
  - Conversion: 1 kilogram (kg) = 2.2 lb
  - 22 lb/2.2 = 10 kg
- Review medication reference to determine recommended range in

Milligrams/kilogram (mg/kg).

- Reference text states recommended dosage range is 30 to 50 mg/kg/day in 4 divided doses.
- Multiply child's weight by the lower and upper limits of the dosage range. • 10 kg × 30 = 300 mg/day
  - 10 kg × 50 = 500 mg/day
- Child's recommended range = 300 to 500 mg/day
- Compare physician's order with dosage range to determine whether medication dosage is safe.
- Order reads: Give 500 mg q6h.
  - 500 mg given q6h = 500 × 4 = 2000 mg/day
- Conclusion: Dosage exceeds recommended range.
- Dosages outside recommended range should be verified with physician. "Dr. Jones, I've just reviewed the new medication order for Sam Lee. According to my calculations, the dosage exceeds that recommended for his weight according to both the formulary and pharmacist."

Example: The child you are caring for has an order for Amoxicillin 400 mg PO BID. Begin by checking the dose against the pediatric drug dosing handbook. The child's weight is 10 kg. You note that the appropriate dose for Amoxicillin is 80 mg/kg/ day divided BID. Set up the equation by multiplying the recommended dose by the child's weight. Divide the dose calculated (80 x 10 = 800) by the number of doses recommended (800 / 2 = 400). So, an appropriate dose for a 10 kg child would be 400 mg per dose, two times per day.

Other examples (Appendix 1).



purpose

The rectal route for administration is less reliable but is sometimes used when the oral route is difficult or contraindicated. It is also used when oral preparations are unsuitable to control Vomiting.

Procedure:

- Remove the wrapping on the suppository and lubricate the suppository with warm water (water-soluble jelly may affect medication absorption).

- Reverse contractions or the pressure gradient of the anal canal may help the suppository slip higher into the canal.

- Using a glove or finger cot, quickly but gently insert the suppository into the rectum beyond both of the rectal sphincters
- Then hold the buttocks together firmly to relieve pressure on the anal sphincter until the urge to expel the suppository has passed, which occurs within 5 to 10 minutes.

- If medication is administered via a retention enema, the same procedure is used. Drugs given by enema are diluted in the smallest amount of solution possible to minimize the likelihood of being evacuated.

I.V. Therapy

Introduction

In children, I.V. therapy may be prescribed to administer medications or to correct a fluid deficit, improve serum electrolyte balance, or provide nourishment. Primary nursing concerns related to pediatric I.V. therapy include correlating the I.V. site and equipment with the reason for therapy and the patient's age, size, and activity level. For example, a foot or scalp vein I.V. site may be used for infants, whereas a peripheral hand or wrist vein may be more suitable for ambulatory older children. During I.V. therapy, the nurse must continually assess the patient and the infusion to prevent fluid overload and other complications.

Equipment

- ❖ Prescribed I.V. fluid
- ❖ volume-control set with micro drip tubing
- ❖ infusion pump
- ❖ I.V. pole
- ❖ normal saline solution or sterile dextrose 5% in water (D5W) for injection
- ❖ povidone-iodine solution
- ❖ alcohol pads
- ❖ 3-ml syringe
- ❖ child-size butterfly needle or I.V. catheter
- ❖ tourniquet • ½″ or 1″
- ❖ sterile tape gloves
- ❖ optional: arm board, insulated foam or medicine cup, luer-lock cap, air eliminator I.V. filter.

Whenever possible, use a catheter instead of a needle. A flexible catheter is less likely to perforate the vein wall. To promote compliance and reduce the discomfort associated with catheter insertion, consider using a transdermal anesthetic cream.

Preparation of equipment

☒ Gather the I.V equipment and take it to the patient's bedside. Check the expiration date on the I.V. fluid and inspect the I.V. container (an I.V. bag for leakage, a bottle for cracks). Examine the I.V. tubing for defects or cracks. Make sure the packaging surrounding the I.V. catheter or needle remains intact.

☒ Open the wrappings on the I.V. solution and the volume-control tubing set. Close all clamps on the tubing set; then insert the tip of the tubing set into the entry port of the I.V. bag or bottle. (If you're using an I.V. bag, be sure to hold the bag upright when attaching the tubing. This will keep the sterile air inside the bag from escaping and making the fluid level difficult to read.)

☒ Hang the bottle or bag from the I.V. pole. Open the clamp between the bag and the volume-control set and allow 30 to 50 ml of solution to flow into the calibrated chamber. Close the clamp.

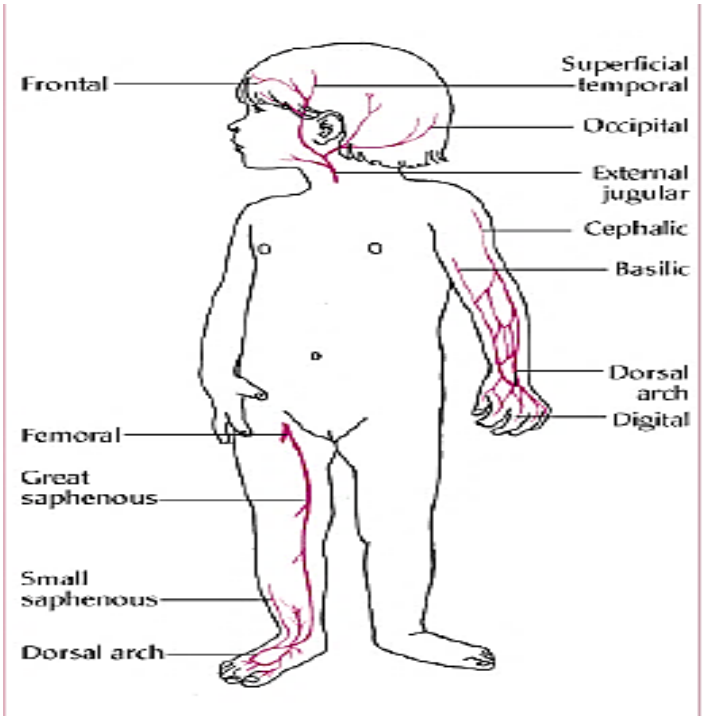
☒ Squeeze the drip chamber located below the calibrated chamber or volume-control set to create a vacuum. Release the drip chamber and allow it to fill halfway with solution. Then release the clamp below the drip chamber so that fluid flows into the remaining tubing, removing any air. After the tubing fills, close the clamp.

TABLE 19-2 Age-Based Guidelines for Selecting Needle Length and Gauge

AGE GROUP	NEEDLE LENGTH	NEEDLE GAUGE
Infant	5/8 inch	25–27
Toddler or Preschool	1 inch	22–23
School-age or adolescent	1–1½ inch	22–23

Common pediatric I.V. sites

Below are the most common sites for I.V. therapy in infants and children. Peripheral hand, wrist, or foot veins are typically used with older children, whereas scalp veins are used with an infant.



Procedure

❋ Match the name on the patient's wristband with the name on the physician's order (or on the medication card). Ask the patient (or his parents) whether he's allergic to povidone-iodine solution or to any type of tape.

❋ Explain the reason for the I.V. therapy. Reassure the parents, and enlist their assistance in explaining the procedure to the patient in terms he can understand.

❋ Be sure to have a staff member available to assist you. Inform the parents that the staff member will help the patient remain still, if necessary, during the procedure.

❋ Wash your hands and put on gloves.

❋ Select the insertion site for the butterfly needle or catheter. (See Common pediatric I.V. sites .) Aim for the most distal site possible, and avoid placing the I.V. line in the patient's dominant arm or in areas of flexion if possible. Avoid previously used or sclerotic veins.

❋ To locate an appropriate scalp vein, carefully palpate the site for arterial pulsations. If you feel these pulsations, select another

site. Then, before inserting the I.V. line, prepare the selected site as ordered.

❋ To find an appropriate peripheral site, apply a tourniquet to the patient's arm or leg and palpate a suitable vein.

❋ If you're inserting a butterfly needle, flush the tubing connected to the butterfly with D5W or normal saline solution.

❋ Clean the insertion site. Unless contraindicated, use a pad containing povidone-iodine solution. Wipe with a circular motion from the insertion site's center to the outer rim. Let the solution dry.

❋ Insert the I.V. needle into the vein. Watch for blood to flow backward through the catheter or butterfly tubing, which confirms that the needle is in the vein.

❋ Loosen the tourniquet, and attach the I.V. tubing to the hub of the needle or catheter. Begin the infusion.

❋ Secure the device by applying a piece of ½″ tape over the hub. Next, place a piece of tape, adhesive side up, underneath and perpendicular to the device. Lift the ends of the tape and crisscross them over the device. Further secure and protect the I.V. line as needed.

NURSING ALERT Remember to use only sterile tape under the transparent dressing.

❋ Adjust the infusion flow, as ordered, by using the clamp on the I.V. volume-control tubing or by setting the infusion rate on the infusion pump.

❋ Add solution hourly (or as needed) from the I.V. bag to the volume-control set.

❋ Assess the I.V. site frequently for signs of infiltration, and check the I.V. bottle or bag for the amount of solution infused.

❋ Change the I.V. dressing every 24 hours, according to your facility's policy, or as needed to prevent infection. Change the I.V. tubing every 48 to 72 hours and the I.V. solution bottle or bag

every 24 hours. Label the I.V. bottle or bag, tubing, and volume control set with the time and date of change.

❋ Change the I.V. insertion site every 72 hours, if possible, to minimize the risk of infection. If you inserted the I.V. line without proper skin preparation (during an emergency, for example), change the site sooner.

Complications of I.V therapy

Infection, fluid overload, electrolyte imbalance, infiltration, and circulatory impairment are complications of I.V. therapy.

Documentation

Record the date and time of the I.V. infusion, the insertion site, and the type and size of I.V. needle or catheter. Note the patient's tolerance of the procedure. Describe patient- and parent-teaching activities. Document the condition of the I.V. site according to hospital policy. If infiltration affects the I.V. site, document the condition of the site at every shift change until the condition resolves.

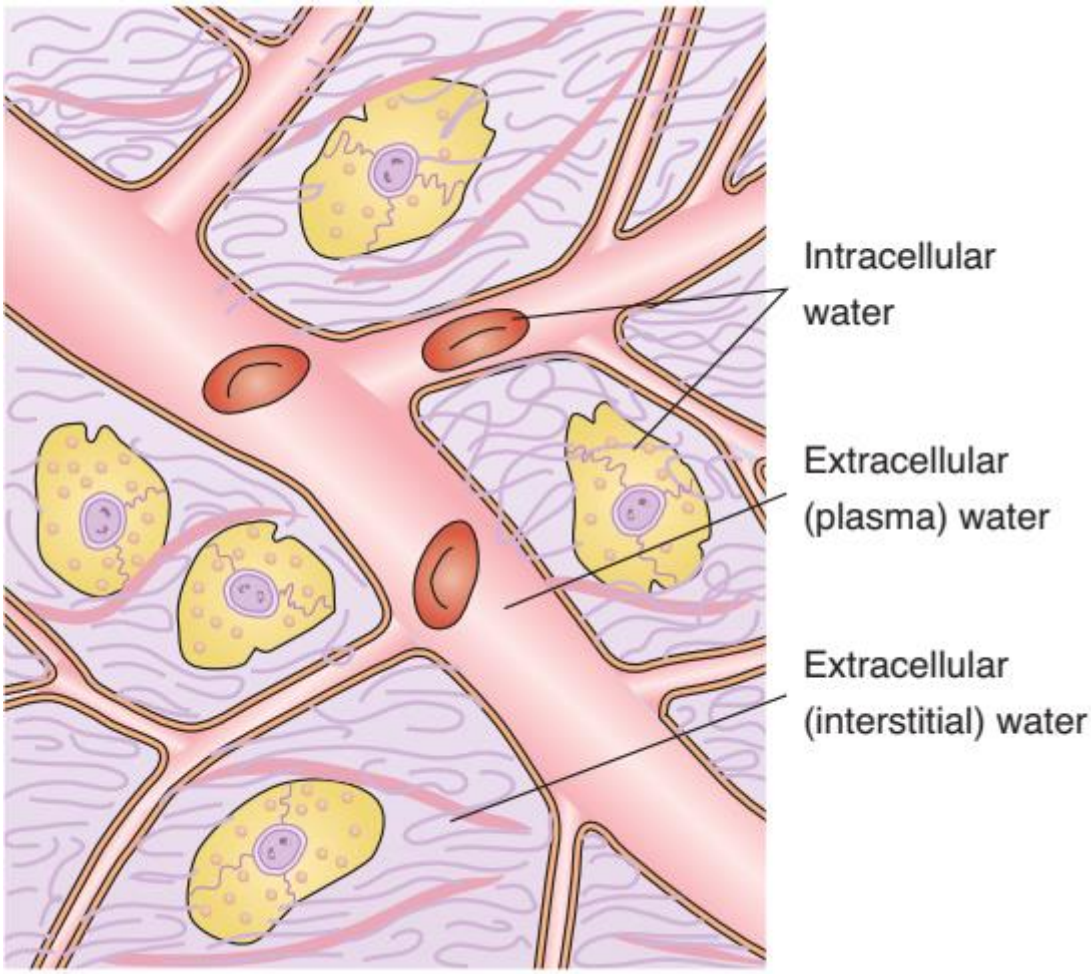
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Fluid Calculation and Calories Requirement

Objective

1. Explaining compartments of body fluid
2. Calculating Pediatric Fluid Needs
3. Describe Types of I.V. solutions
4. Identify calories required for child
5. Calculating a Formula's Adequacy



Body fluid composition

Body fluids are separated into two main compartments:

1. **Intracellular** (ICF 40%)
2. **Extracellular** (ECF) (divide into intravascular 5% and interstitial fluid 25%)

- The proportion of body weight that is water falls from about 78% in a term newborn to 60% in adults.
- Water balance depends on **intake, output and usage for metabolism**. Water is normally lost from skin, lung, intestine and kidney.
- Cell membranes are **relatively permeable** to water, potassium and chloride, and **relatively impermeable** to sodium, phosphate and protein. Sodium is actively transported out of cells by energy-dependent sodium pumps.
- There is equilibrium in tonicity, hydrostatic and colloid osmotic pressure between plasma and interstitial fluid. Water leaves the arterial end of the capillary under hydrostatic pressure and is drawn into the venous end of capillary beds by plasma oncotic pressure.
- The plasma osmolality almost constant between 285 and 300mOsmol per kg H2O.

Calculating Pediatric Fluid Needs

Maintenance requirements involve replacement of normal losses from urine, sweat, lungs and feces, and assume that any fluid deficit (dehydration) has been replaced.

Fluid needs based on weight

The usual daily maintenance requirement can be calculated from the following formula: (South & Isaacs, 2012)

1. Children weighing under 10 kg require 100 ml of fluid per kilogram of body weight per day:

   Weight in kg ×100 ml/kg/day = fluid needs in ml/day

   e.g. child weight 7Kg => 7 × 100 ml = 700ml/day

2. Children weighing 10 to 20 kg require 1,000 ml of fluid per day for the first 10 kg plus 50 ml for every kilogram over 10:

11–20kg: 1000mL + 50mL/kg for every kilogram above 10kg= fluid needs in ml/day

e.g. child weight 15Kg => 1000ml +50ml × 5kg = 1250ml/day

3. Children weighing more than 20 kg require 1,500 ml of fluid for the first 20 kg plus 20 ml for each additional kilogram:

Above 20 kg: 1500mL + 20 mL/kg for every kilogram above 20 kg= fluid in ml/day

e.g. child weight 24Kg => 1500ml +20ml × 4kg = 1580 ml/day

The hourly maintenance requirement can be calculated from the following formula:

1. First 10 kg body weight: 4mL/kg per h.
2. 11–20 kg: (4 mL/kg for the first 10kg = 40 mL) + 2mL/kg for every kilogram above 10kg.
3. Above 20 kg: (4 mL/kg for the first 10kg = 40mL) + (2mL/kg for the second 10kg = 20 mL) + 1mL/kg for every kilogram above 20kg.

Table 6.1.5 Daily and hourly maintenance fluid requirements

Weight	Maintenance fluid requirement for 24 h	Maintenance fluid requirement per hour
First 10 kg	100 mL/kg	4 mL/kg
Second 10 kg	50 mL/kg	2 mL/kg
Subsequent kilograms	20 mL/kg	1 mL/kg

The volume of fluid deficit may then be calculated using the following formula:

Calculated fluid deficit (mL) = %dehydration ×weight (kg) × 10

(e.g. 7% dehydration in a 12 kg child = 7 × 12 × 10 = 840 mL deficit).

Types of I.V. solutions

Solutions used for I.V. fluid replacement fall into the broad categories of **crystalloids** (which may be isotonic, hypotonic, or hypertonic) and **colloids** (which are always hypertonic) (Lippincott, 2012).

A. **Crystalloids** are solutions with small molecules that flow easily from the bloodstream into cells and tissues.

Crystalloids I.V. solutions

Isotonic
Isotonic solutions expand the intravascular compartment. Isotonic solutions include:
(D5W, 0.9% NaCl, Ringer's solution and lactated Ringer's solution).

Hypertonic
Hypertonic solutions greatly expand the intravascular compartment and draw fluid from intravascular areas. Hypertonic solutions include: (D10W, 3% NaCl, 5% NaCl, D5 0.45% NaCl and D5 0.9% NaCl).

Hypotonic
Hypotonic solutions cause a fluid shift from the intravascular compartment into the cells. Hypotonic solutions include: (D2.5W, 0.45% NaCl and 0.33% NaCl)

B. Colloids

The use of colloids over crystalloids is controversial. Still, the doctor may prescribe a colloid—or plasma expander—if your patient's blood volume doesn't improve with crystalloids. Examples of colloids that may be given include:

• albumin (available in 5% solutions, which are osmotically equal to plasma, and 25% solutions, which draw about four times their volume in interstitial fluid into the circulation within 15 minutes of administration)

• Plasma protein fraction

• Dextran

Energy / Calories requirement

- Food is metabolized and provides energy required by the body for growth and synthesis of new tissue, for metabolic processes, and for physiological functions and activity.

- Fat, carbohydrate and protein provide energy, which is measured in kilocalories. Fat contributes 50% of the total energy in breast milk or standard infant formula. The average diet of older children provides 30–40% of total energy from fat, 45–55% from carbohydrate and about 15% from protein (South & Isaacs, 2012).
- On average, carbohydrates and protein contain 4 calories per gram, fats contain 9 calories per gram

An infant (birth-6 months) of age requires 110 to 120 calories per kg /day.

- After 6 months of age, this amount gradually declines until the requirement at 1 year is 100 kcal/kg per day. In contrast, the adult requirement is 42 kcal/kg per day.
- Adolescents have an increased caloric need due to rapid growth. For most adolescent boys, 2,800 calories per day & girls, 2,200 kcal/kg/day is recommended (AAP, 2011)
- Caloric Requirements (Approximate)
- Preterm Infants 120 Kcal/kg/day
- Infants 100 Kcal/kg/day
- Young Child 80 Kcal/kg/day (1-10year)
- Older Child 60 Kcal/kg/day

There are three type of milk each cup (200ml) contain:

1. Whole fat milk (8gm fat , 12gm carbohydrate and 8gm protein)= 150kcal
2. Low fat milk (5gm fat , 12gm carbohydrate and 8gm protein)= 120kcal
3. Free fat milk (skimmed) (1gm fat , 12gm carbohydrate and 8gm protein)= 90kcal

- skim milk and low-fat milk must not be used for infant because they are deficient in caloric concentration, significantly increase the renal solute load and water demands, and deprive the body of essential fatty acids (Hockenberry & Wilson, 2015).
- Low-fat (2%) cow's milk can be given to children over 2 years of age. Skim milk (no fat) is not recommended unless there is a specific medical condition, such as hypercholesterolemia.
- Whole Cow's milk is appropriate to be introduced to infants over 12 months.
- Cow's milk has lower iron content and higher levels of protein, sodium, potassium, phosphorus and calcium than human milk or formula therefore it's not recommended for less than 1 year child.

Calculating a Formula's Adequacy

- A quick rule of thumb to estimate how much an infant will drink at a feeding is to add 2 or 3 to the infant's age in months. After initially taking 0.5 to 1 oz for the first 2 days, a newborn (0 age) will take 2 to 3 oz each feeding; a 3-month-old child, 5 to 6 oz (Pillitteri, 2010).

* Both commercial formulas and breast milk contain 20 calories/oz.

*The total fluid ingested for the infant: 75 to 90 mL (2.5 to 3 oz) / lb per day.

*1 kg = 2.2 lb

e.g. infant (10kg=22 lb) take 55-66 oz => (55-66 oz/day × 20kcal = 1100-1320kcal/day)

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Oxygen Therapy

Objective:

By the end of this chapter, the student nurse will be able to:

1- Define oxygen therapy.
2- List the purposes of oxygen therapy.
3- Identify indications of oxygen therapy.
4- List different methods of oxygen therapy.
5- List the equipment required for oxygen therapy.
6- Perform an appropriate technique for each method.

Introduction:

Oxygen is an essential body requirement for any energy consumption, activity or function. Hypoxemia resulting from apnea or inadequate ventilation and it occurs more rapidly in children than in adults because of children's higher metabolic rate and increase oxygen consumption.

Definition:

It is a type of treatment given to patient with acute or chronic respiratory diseases or ventilation failure as a lifesaving.

Purpose:

1. To prevent hypoxemia by adequate transport of oxygen in blood in patients with pulmonary diseases

2. Provide adequate tissue oxygenation for problems associated with

3. Reduced oxygen carrying capacity of blood

4. Decreased cardiac output

5. Hypoventilation

6. Increased metabolism

Indication:

1- Decrease environmental oxygen:

- Inhalation of non-physical gas mixture.

2- Impaired alveolar ventilation:

- Severe airway obstruction (upper or lower).
- Severe lung pathology (pneumonia, pulmonary edema).
- Severe respiratory weakness (C.N.S depression, respiratory muscles paralysis).

3- Abnormal pulmonary blood flow:

- Persistant fetal circulation.
- Acute pulmonary hypertension.

4- Decreased oxygen carrying capacity (low hemoglobin):

- Acute haemolytic anemia.
- Acute blood loss.

5- Poor tissue perfusion:

- Shock state.

6- Acute brain insult:

- Convulsion – coma – increase intracranial pressure.

Types:

1- High flow rate	2- Low flow rate
- 13-15L/min.	- 1-13L/min.
- 60-80%	- 40-60%

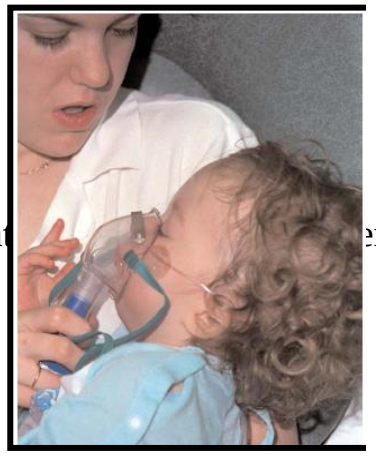
Methods:

1- Simple face mask.
2- Nasal cannula.
3- Hood (head box).
4- Mist tent.
5- Incubator.
6- Ambo bag.
7- Venturi mask (T. tube).

Table 16.1: Determining oxygen percentages by various devices

	Liters	*Nasal cannula*	*Simple mask*	*Partial rebreather mask*	*Non-rebreather mask*
	2	23% to 28%	Not applicable	Not applicable	Not applicable
	3	28% to 30%	— do—	— do—	— do—
	4	Not applicable	— do—	— do—	— do—
Oxygen delivery	5	— do—	40%	— do—	— do—
rate liters/min	6	— do—	45% to 50%	35%	55% to 60%
	8	— do—	Not applicable	45% to 50%	60% to 80%
	10	— do—	— do—	60%	80% to 90%
	12	— do—	— do—	60%	90%
	15	— do—	— do—	60%	90%

1- Simple face mask



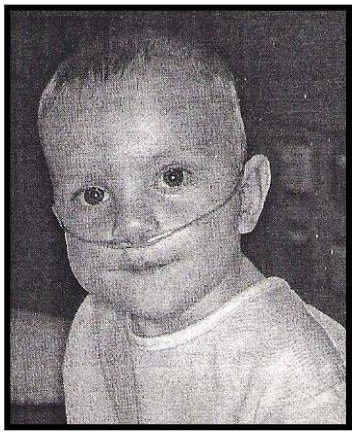
Advantages:

- Available in wide variety of size.
- Suitable for all ages.
- Maximum flow rate 5-10L/m.
- Ability to provide predictable concentrat…er child breathes through nose or mouth.

Disadvantage:

- Child may not tolerate the mask.
- Not suitable for long term therapy.
- The O_2 concentration is depend on the security of…ner face mask.
- Eating talking disrupts oxygen delivery.

2- Nasal Cannula



Advantages:

- Suitable for all ages.
- Prongs allow considerable mobility.
- Doesn't distort child visual field.
- Flow rate 1-5L/m.

- Provision of constant O_2 flow even while child eats or talks.

Disadvantages:

- Young children often resist cannula.
- May irritate nasal mucosa.
- Restrain may be needed.
- Difficulty of controlling O_2 concentration if child breaths through mouth.

3- Head box (hood)

Advantages:

- Available at any size.
- Maximum flow rate (10L/m).
- Facial movement isn't restricted.

Disadvantages:

- Limit access to the head.
- High humidity environment.
- Need to remove box during feeding and providing care to patient.

4- Mist tent

Advantages:

- Tent allows older children considerable mobility.
- Maximum flow rate (10L/m).

Disadvantages:

- Steady O_2 concentration difficult to maintain.
- Child feel isolated.
- Cool and wet tent environment.
- Poor access to patient.

5- Incubator

Advantages:

- Only suitable for neonate and young infant.
- Mobility isn't restricted.
- Infant is easily observed for assessment.
- Infant is easily accessible for intervention.

Disadvantages:

- An O_2 hood must be placed inside it to achieve concentration higher than 85%.

Equipment:

- O_2 source (cylinder or central piping system).
- Delivery equipment.
- Oxygen flowmeter.
- Oxygen tubing.
- Humidification attachment.
- No smoking sign.
- Disposable gloves.
- Water soluble lubricant (for catheter insertion).

Humidification

As oxygen and other compressed gases are dry (contains no water vapor) humidity is added before inhalation of the gas. Humidifiers are devices that add water vapor to inspired gas to decrease drying effect of the gas. Bubbling the O2 through water should humidify oxygen administration. The water may be at room temperature.

Advantages of Humidification

- Prevents drying and irritation of respiratory mucous membranes.
- Prevents drying and thickening of respiratory tract secretions.
- Loosens secretions.

Procedures of nasal cannula, mask or hood

Steps:

1- Perform hand hygiene.
   R. *Reduces transmission of microorganism.*
2- Select proper size of cannula or mask.
   R. *improper size can lead to nasal obstruction.*
3- Remove all friction toys form the area and display no smoking signs.
4- Connect the flow meter to either oxygen source wall unit or freestanding tank.
   R. *Allow for the oxygen to flow from at the prescribed rate.*
5- Connect the humidifier to oxygen setup.
   R. *Provides for moisture in system.*

Humidified air is less drying to nares and lungs, prevent dehydration of mucous membrane and liquefaction of thick mucus

6- Following instruction for particular oxygen setup and fill reservoir with sterile water.

R. *Use sterile water decreases the incidence of bacterial growth.*

7- Attach tubing o oxygen source.

R. *Allow oxygen to flow from source to child.*

8- Connect the distal of oxygen tubing to delivery device (cannula, mask or hood).

9- Turn on flowmeter to the prescribed amount.

R. *To ensure that child receive amount of oxygen physician deemed necessary.*

10- Feel oxygen flowing through the system.

11- Place child in supine semi fowler's position.

R. *Raising the head of the bed protect the airway if the child should vomit during air way placement.*

N.B:

For nasal cannula:

12- Place nasal prong inside external meatus of the nares.

13- Instruct child to breathe through his or her nose.

R. *Provides the prescribed oxygen.*

For mask:

14- Place the oxygen mask over the mouth and nose.

15- Tighten the straps attached to the mask until you can easily fit one finger between the strap and the face of the child.

R. *A properly secured mask allow the prescribed amount oxygen to flow into respiratory system.*

For oxygen hood:

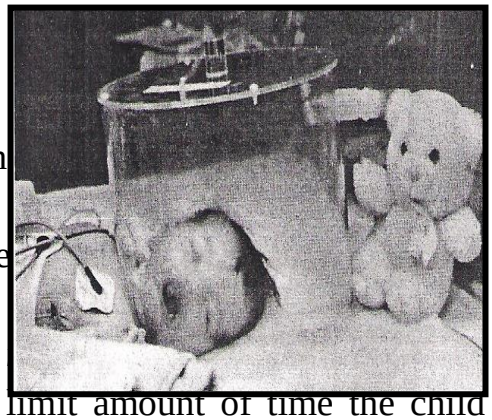
16- Connect the unit to oxygen source.

17- Place the hood on the crib or bed so th

R. *Prevents obstruction of the airway.*

18- The hood should not rest on child's ne to turn or her head side to side easily.

R. *An improperly sized hood can result*



19- Encourage family and other staff to limit amount of time the child is outside of hood.

R. *To maximize the quality of oxygen therapy.*

Evaluation of response:

- O_2 saturation (SaO_2) by pulse oximeter.
- PaO_2 by A.B.Gs.
- If there is good response:
  - SaO_2 will be above 90% or PaO_2 will be above 90 mmHg.
- If there is poor response:
  - SaO_2 will be under 85% or PaO_2 will be under 60 mmHg.

Complications:

The most important complications are:

- Retinopathy of prematurity
- Hypoventilation
- Absorption atelectasis
- Bronchopulmonary dysplasia.

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Immunization and vaccination

Learning objectives

1. To know the Iraqi immunization schedule
2. To understand the details of each vaccine (type, route, doses, side effects)
3. To recognize the differences between two types of polio vaccine
4. To understand what to check before giving vaccine and general instructions about vaccination

Immunization

Immunization is defined as the procedure by which the body is prepared to fight against a specific disease. It is used to induce the immune resistance of the body to a specific disease. **Types of immunization**

I- Passive immunization

It is classified into:

A. Natural Passive Immunization	**B. Artificial Passive immunization:**

• It includes passage of maternal antibodies to the fetus through the placenta. • These antibodies disappear from the infant circulation at about the fourth month of age. (e.g., antibodies against measles, poliomyelitis, diphtheria, scarlet fever and mumps).	• It is an administration of immunoglobulin (Gamma globulins) which contains antibodies it may be antiviral, antibacterial, or antitoxins either alone or combined. • It has short duration. • it is used in emergency cases and for contacts.

II- Active immunization

It is an entrance of antigens, which stimulates formation of antibodies by the immune system. It is classified into:

A. Natural Active immunization	**B. Acquired Active immunization**
• By acquiring the infections either sub - clinically or clinically. • It is life long in measles, German measles, chicken pox and mumps.	• By vaccines

Antibody: a protein that is produced in response to a particular pathogen.

Antigen: the substance that induces the production of antibodies.

Vaccine

It is a suspension of live or killed microorganisms are antigenic portion these agents presented to potential host to induce immunity to specific disease cause by that organism.

Toxoid: A modified bacterial toxin that is made nontoxic but is still able to induce an active immune response against the toxin. and toxoids (tetanus and diphtheria)

The term **immunity** refers to all the processes used by the body to protect against foreign material from environmental sources, including microorganisms or their toxins, foods, chemicals, pollen, dander, or drugs.

Types of vaccine

- Live attenuated (Sabin (polio)- Measles–Mumps– BCG)
- Inactivated or killed (Salk(polio)– Influenza- Rabies).
- Cellular fraction (Pnumoccocal Pneumonia – Meningococcal).

- Recombined vaccine (hepatitis B)
- Toxoids (Tetanus, Diphtheria).
- Immunoglobulin (Hepatoglobulin)
- Antitoxin (Tetaglobulin)

The new recommended schedule of immunization in Iraq (2018)

Age	Vaccine	Route	Dose
At birth	HVB1+ BCG+ OPV(0)	IM+ ID+ oral	**0.5ml, 0.05ml**
2 Months	Hexa(1)+POLIO(1)+Rota(1) +PCV1	IM+ORAL+ORAL +IM or SC	**0.5ml, 2ml**
6 Months	Hexa(3)+POLIO(3) +PCV3	IM+ORAL	**0.5cc**
9 Months	MEASLES+VIT A(100000IU)	SC+ORAL	**0.5cc**
15Months	M.M.R	SC	**0.5cc**
18-24 Months	FIRST.B00STER POLIO+ Penta + 200000IU VIT. A	ORAL+IM+ORAL	**0.5cc**
4-6 Years	2nd..B00STER POLIO+ Tetra+ MMR	ORAL+IM+SC	**0.5cc**

Abbreviations with meaning

Hexa:(DTaP, Hib, HBV and IPV)
Penta (DTaP, Hib and IPV)
Tetra (DTaP and IPV)
ROV=Rota virus vaccine = severe diarrhea
DTP=diphtheria, tetanus, pertussis
IPV= inactive or injection polio vaccine
OPV= oral polio vaccine= poliomyelitis
Hib=Haemophilus influenza type B
MMR= measles, rubella, mumps
PCV = pneumococcal conjugate vaccine, streptococcal pneumonia
BCG=Bacille Calmette-Guerin, Tuberculosis
* If the infant develops a severe reaction to a prior dose, give DT not DPT, because the pertussis component is responsible for this severe reaction.

BCG: contraindication with side effects

- The dose is 0.05ml intradermal in the deltoid region, successful vaccine produces a small indurated area (2-4mm) after 3-4weeks. The lesion progresses to a papule or shallow ulcer of approximately 10 mm diameter and heals within 12 weeks to form
- small, flat scar.
- If given subcutaneously (instead of the usual intra-dermal route) it will lead to abscess formation & regional lymphadenopathy.
- If the child is Tuberculin positive, it will lead to a severe reaction.
- Efficacy 50%for pulmonary TB, 50-80%for disseminated & meningeal TB.

B.C.G Side effects:

- Regional Lymphadenitis.
- Abscess.
- Osteomyelitis.
- Disseminated B.C.G infection (if immune deficient)

Poliomyelitis Vaccines

1. Polio Vaccine

There are two types of polio vaccine Sabin and Salk vaccine.

A: sabin (oral) vaccine:

- It is preferred agent for prophylaxis.
- It eases of administration.
- It is a vaccine containing live attenuated viruses.
- The vaccine is clear pink liquid and is kept at 00C to +80C at the Health Center on the top shelf of the refrigerator.
- Keeps the vaccine in a cup containing ice during the vaccination session.
- This immunization protects not only the individual but also gives mass immunization to the community because the immune person is less likely carry the wild virus in his intestine and acts as a vector to the others due to the gained gastrointestinal immunity.
- Throw away the vaccine that remains in the vial after every vaccination session even if this vial was opened for a single child, and do not keep it in the refrigerator.
- Let the mother hold the infant lying on his back and open the infant mouth by squeezing cheeks gently between your fingers then put two drops of the vaccine straight from the dropper into the child tongue.
- Repeat the dose if the child spits out the drops.

- Lactation has no effect on the potency of the vaccine.
- The vaccination may be not effective if the child has diarrhoea. So, after giving the vaccine it must be repeated. There is no harm in repeating the vaccination

B: Salk vaccine (inactivated vaccine)

- Effective in prevention of polio.
- The vaccine has longer life span than Sabin.

It has to be injected so, it is difficult to be used in mass immunization

DPT vaccine: side effects

Side effects:

The side effects are due to the pertussis and include:

- **Mild side effects:** Local swelling and tenderness, slight fever but fever may reach to 40 ^{o}c and disappears in few days.
- **Moderate side effects:** shock
- **Severe side effects:** Convulsions, encephalopathy, and brain damage but are rare.

NB:

1 If one of severe complications occurred do not give the pertussis again, but we will give D.T vaccine only.

2 Pertussis vaccine is not given after 6 years of age for the fear of complications so D.T is given.

Minor reactions are quite frequent in 20–50% of vaccines. Local reactions include: Inflammation, induration or a painless nodule at the site of injection. These are progressively more common after the first injection

– Moderate reaction occur in 0.1% to 1.0% of children and include:

- ongoing crying (for three hours or more in the first 12 hours),
- a high fever (up to 40°C), and
- an unusual (screaming), high-pitched crying

– Severe problems happen very rarely (1 in 140,000 doses of DPT). Include;

- a serious allergic reaction,
- prolonged seizures,
- encephalopathy, or even death

MMR vaccine: side effects

Because MMR is a live-attenuated vaccine, non–allergy-related side effects are noted 5 to 12 days following immunization.

- Fever and rash are relatively common, experienced by 5% to 15% of recipients.
- Transient arthritis has been reported.
- Thrombocytopenia (rare)
- Encephalopathy (very rare)
- A general rule of thumb is the "rule of 10"—about 10% of children get a rash approximately 10 days after vaccine administration.

MMR: Contraindications

- Severe allergic reaction to vaccine component or following prior dose
- Pregnancy
- Immunosuppression
- Moderate or severe acute illness
- Recent blood product

Measles side effects:

Side Effects

- Mild fever and rash develop after 8-12 days, and last for 1-3 days in about 15% of vaccinated children.
- The risk of encephalitis 1/106 but the risk of developing encephalitis as a complication of measles is 1/1000 so by vaccinating children, we have diluted the incidence of encephalitis by 1000 times.
- Other complications of measles are diarrhoea, bronchitis, pneumonia, conjunctivitis and malnutrition. These are also prevented by vaccination.

Hepatitis B vaccine side effects:

Transient erythema and induration at the site of injection, fever malaise, flu-like illness, arthritis, myalgia and arthralgia.

Rota virus vaccine: side effects

diarrhea, vomiting, Nasopharyngitis, otitis media, and bronchospasm.

Immuno-compromised children:

- In general, children who are immunological compromised should not receive live bacterial or viral vaccines.

Immunodeficiency:

- Live-bacteria (e.g., oral typhoid) and live-virus vaccines (e.g., MMR, varicella, and rotavirus) are contraindicated in most circumstances involving clinically significant immunosuppression
- Vaccines should be postponed until 3 months after immunosuppression drugs have been stopped.

Nursing actions before vaccine administration

Be sure to ask these questions before giving the vaccines:

- ✓ Is your child sick today? (More than a common cold, earache, etc.)
- ✓ Does your child have any severe (life-threatening) allergies?
- ✓ Has your child ever had a severe reaction after a vaccination?
- ✓ Does your child have a weakened immune system (because of diseases such as cancer, or medications such as steroids)?
- ✓ Has your child gotten a transfusion, or any other blood product, recently?
- ✓ Has your child ever had convulsions or any kind of nervous system problem?

7. Does your child not seem to be developing normally?

Care minder

High fevers and severe illness are reasons to delay immunization, but only until the child has recovered from the acute stage of the illness to avoid adding the risk of adverse side effects from the vaccine to a child who is already ill, or confusing symptoms of the illness with a side effect of the vaccine. Minor illnesses such as a cold, otitis media, or mild diarrhea without fever are not contraindications to immunization.

Potential Reactions Associated with Immunization and intervention

Potential reaction	Nursing intervention
Pain with injection	Apply topical anesthetics at the injection sites, giving enough time for the anesthetic to have the desired effect. Administer oral sucrose for infants up to 3 months of age. Encourage parents to hold a young child during injection. Use distractions such as bubbles, blowing on pinwheels, and reading stories to reduce attention to pain and potentially relax the muscle. Encourage teens to use headsets to listen to music.
Fever **Myalgia**	Acetaminophen as above

Nurses should answer mother's questions regarding vaccination

1. **What are Immunizations?** Vaccines act by stimulating the body to make defenses against the infection

2. **Are childhood immunizations safe?** In general, most babies and toddlers do not have reactions to immunizations. If they do, the reactions are usually mild and of no long-term significance.
3. **What is the point of immunizing my baby against diseases that are not common where I live?** If the majority of people stop immunizing their children, we could be in a situation like Ethiopia where measles becomes a killer disease again.
4. **Can my child have immunizations if he is allergic to eggs?** Children who are allergic to eggs can have all the usual childhood immunizations. The only immunization that should be avoided if your child is allergic to egg is influenza vaccine.
5. **Should I delay immunizing my child if he is unwell?** If your child has a mild illness (say a runny nose) and doesn't have a fever, he can be immunized. Young children have so many viral infections (see Upper Respiratory Tract Infections), that to delay immunization for these would mean your child would not be protected from more serious infections.
6. **When should I start immunizations if my baby was born early?** Even if your baby is premature, the infant immunizations start at the baby's chronological age, not their gestational age.

Storage and transportation of vaccine:

The cold chain and vaccine supply and handling:

WHO and UNICEF have adopted standards for the safe shipping of vaccines. These require that vaccines be properly insulated during international shipment advanced notice of shipment is sent to receiving authorities and a time/ temperature monitor (called a cold chain monitor) is included with each vaccines.

When the vaccine is exposed to a temperature above 10^0C, a blue color appears on the monitor, indicating a cold chain break. The monitor also provides an indication of the extent of the break and what to do.

Ensuring that no breaks occur in the cold chain requires continuous monitoring of the transport and storage condition of vaccines.

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2-Baiee , H. 2012. The Expanded Program on Immunization. Available from: http://www.uobabylon.edu.iq/uobColeges/lecture.aspx?fid=15&depid=1&lcid=24463

3- World Health Organization(WHO), 2019. Available from: https://www.who.int/topics/immunization/en/

4-Website :https://wikieducator.org/Lesson_9:_Immunization

Nasogastric Intubation

Objectives:

By the end of this procedure, the student will be able to:

1- Define Gavage feeding.
2- List indications of Gavage feeding.
3- Enumerate purposes of Gavage feeding.
4- Recognize criteria of tube selection.
5- Identify equipment needed for Gavage feeding.
6- Demonstrate steps of Gavage feeding.
7- Recognize complications of Gavage feeding.
8- List contraindication of Gavage feeding.

Definition of Gavage feeding:

It is a type of enteral feeding which is given to the infant using a nasogastric tube which is a narrow bore tube passed into the stomach via the nose.

Indications:

1. It is used for short or medium term nutritional support.
2. Aspiration of stomach contents, e.g. for decompression.
3. Administration of drugs.
4. Poor sucking and swallowing reflexes.
5. To provide nutrition post operatively.
6. Respiratory distress.
7. Loss of consciousness.
8. Cyanosis after feeding by bottle or medicine dropper.
9. Congenital abnormalities, as cleft lip and cleft palate.
10. Neurologic impairment.
11. A gestational age of less than 34 w.
12. A weight of less than 1500gms.

Purpose of gavage feeding:

1- To provide safe method of feeding or administrating medication where the infant is unable to suck or swallow.
2- To provide a route that allows adequate caloric or fluid intake.
3- To provide safe method of feeding in sick infant.
4- To prevent fatigue or cyanosis which could occur from bottle feeding.

Purpose of nasogastric tube insertion:

A nasogastric tube may be used to decompress the stomach, instill medications or feedings or to assess gastrointestinal function.

Guidelines of nasogastric tube selection:

Factors influencing the selection of feeding tubes include:

- Child's age and weight.
- The viscosity of the formula.
- The reason for the enteral feeding.
- Whether an infusion device will be used.
- A feeding tube (catheter) of size 5 to 10 French is used in infants and the size increases proportionality for the older children.
- Selecting the smallest bore tube for the infusion and a tube of soft material will decrease the child's discomfort.

Type of tube:

- For gavage or lavage, use a single lumen tube.
- For intermittent gastric decompression, use a double lumen tube.
- For continuous long term feeding, use a silicone tube.

Tube size:

Infant / Child	Tube size
2 kg	6 French
3-9 kg	8 French
10-20 kg	10 French
20-30 kg	12 French
30-50 kg	14 French
> 50 kg	16 French

Equipment:

- Sterile catheter.
- Gauze if needed.
- Stethoscope.
- Water for rinsing if needed.
- Clamp.
- Tape, type.
- Non sterile gloves.
- Formula.
- Water for lubrication or other lubricant.
- Sterile syringe (5cc or 10cc).
- Towel and bib.
- Kidney tray.
- Pacifier if necessary.

Performing Gavage Feeding Nursing Action Preparation

■ When choosing the catheter size, consider the size of the infant, the area of insertion (oral or nasal), and the desired rate of flow.

Rationale: The size of the catheter will influence the rate of flow.

■ Explain the procedure to the parents.
■ Elevate the head of the bed and position the infant on the back or side to allow easy passage of the tube.
■ Measure the distance from the tip of the ear to the nose to the xiphoid process, and mark the point with a small piece of paper tape (Figure 28–6) to ensure enough tubing to enter the stomach.



Figure 28–6 Measuring gavage tube length.

Procedure:
Clean Gloves
Inserting and Checking Placement of Tube

1. If inserting the tube nasally, lubricate the tip in a cup of sterile water. Use water instead of an oil-based lubricant, in case the tube is inadvertently passed into a lung. Shake any excess drops to prevent aspiration.
2. If inserting the tube orally, the oral secretions are enough to lubricate the tube adequately.
3. Stabilize the infant's head with one hand and pass the tube via the mouth (or nose) into the stomach to the point previously marked. If the infant begins coughing or choking or becomes cyanotic or phonic, remove the tube immediately as the tube has probably entered the trachea.
4. If respiratory distress is not apparent, lightly tape the tube in position, draw up 0.5 to 1.0 ml of air in the syringe, and connect the syringe to the tubing. Place the stethoscope over the epigastrium and briskly inject the air (Figure 28–7). You will hear a sudden rush as the air enters the stomach.
5. Aspirate the stomach contents with the syringe, and note the amount, color, and consistency to evaluate the infant's feeding tolerance. Return the residual to the stomach unless you are requested to discard it. It is usually not discarded because of the potential for electrolyte imbalance.



Figure 28–7 Auscultation for placement of gavage tube.

Administering the Feeding

1. Hold the infant for feeding, or position the infant on the right side.
Rationale: This position decreases the risk of aspiration in case of emesis during feeding.
2. Separate the syringe from the tube, remove the plunger from the barrel, reconnect the barrel to the tube, and pour the formula into the syringe.
3. Elevate the syringe 6 to 8 inches over the infant's head, and allow the formula to flow by gravity at a slow, even rate.
You may need to initiate the flow of formula by inserting the plunger of the syringe into the barrel just until you see formula enter the feeding tube. Do not use pressure.
4. Regulate the rate to prevent sudden stomach distention leading to vomiting and aspiration. Continue adding formula to the syringe until the infant has absorbed the desired volume.

Clearing and Removing the Tube

1. Clear the tubing with 2 to 3 ml of air. *Rationale: This ensures that the infant has received all of the formula. If the tube is going to be left in place, clearing it will decrease the risk of occlusion and bacterial growth in the tube.*
2. To remove the tube, loosen the tape, fold the tube over on itself, and quickly withdraw the tube in one smooth motion to minimize the potential for fluid aspiration as the tube passes the epiglottis. If the tube is to be left in, position it so that the infant is unable to remove it. Replace the tube per hospital policy.

Maximize the Feeding Pleasure of the Infant

1. Whenever possible, hold the infant during gavage feeding. If it is too awkward to hold the infant during feeding, be sure to take time for holding after the feeding.

Rationale: *Feeding time is important to the infant's tactile sensory input.*
2. Offer a pacifier to the infant during the feeding.
Rationale*: Sucking during feeding comforts and relaxes the infant, making the formula flow more easily. Infants can lose their sucking reflexes when fed by gavage for long periods.*
3. Document procedure, including untoward complications, in the baby's bedside chart or computerized electronic chart.

Documentation:

After insertion of enteral tube, document the following:

- Date and time of insertion
- Type, size\ length tube that used
- Verification method

Complications:

1. Nose bleeding.
2. Sinusitis.
3. A sore throat.
4. Erosion of the nose.
5. Esophageal perforation.
6. Pulmonary aspiration.
7. A collapsed lung.
8. Intracranial placement of the catheter.

References:

1-Tiedeman, M.E. and Mills, D. A., 2002. Pediatric nursing skills. Delmar , a division of Thomson learning, Inc.

2- London, M.L. and Davidson, M.R., 2010. Contemporary maternal-newborn nursing care. Pearson.

3-Bowden, V.R. and Greenberg, C.S., 2011. *Pediatric nursing procedures*. Lippincott Williams & Wilkins.

First aid for the choking child or baby and CPR

Learning objectives

On completion of this lecture students will be able to:

1. Assess the choking child to give urgent intervention.
2. Describe emergent procedures for the relief of foreign body obstruction in an infant or child.
3. Assess the unconscious infant or child.
4. Learn how to perform cardiopulmonary resuscitation for infant and child.

First aid for the choking child or baby and CPR

Choking can be broken up into three sections

1-**If they are unconscious**, we go through DRSABCE

2-**If they have an effective cough**, where they can breathe in and

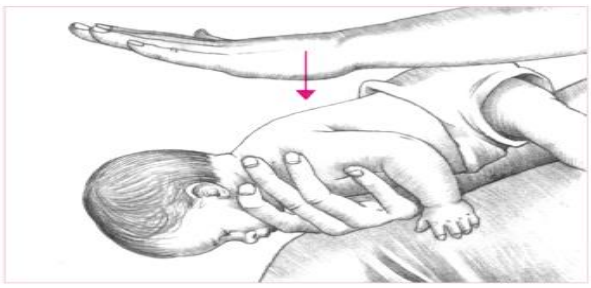


breath out, look in the baby mouth, If the object can be got up, do it but it is right to the back Do not use a blind finger-sweep to discover or remove an obstruction. In an infant, this maneuver may push the object back into the airway and cause further obstruction. Only place your fingers in the infant's mouth to remove an object you can see. We just need to encourage them to cough until the object is out.

3-If they have an in effective cough or silent we should intervene by:

*remove an airway obstruction, place the infant face down on your forearm, with his head lower than his trunk. Support your forearm on your thigh.

• Use the heel of your free hand to deliver five blows between the infant's shoulder blades (as shown below). Back blows are safer than abdominal thrusts in infants because of the size of the infant's liver, the nearness of vital organs, and the poor abdominal muscle tone.



• If the airway remains obstructed, sandwich the infant between your hands and forearms and flip him over onto his back (as shown above right).

• Keeping the infant's head lower than his trunk, give five midsternal chest thrusts, using your middle and ring fingers only, to raise intrathoracic pressure enough to force a cough that will expel the obstruction. Remember to hold the infant's head firmly to avoid injury.

• Repeat this sequence until the obstruction is dislodged or the infant loses consciousness.

• When the foreign object is removed, assess respirations and pulse. Continue revival efforts if needed.

Summary

If they are unconscious perform DRSABC and call for help
If they have an effective cough encourage them to keep coughing
If they have an ineffective cough or are silent perform up to 5 back blows followed by 5 chest thrusts until they object comes out or they become unconscious

DRS ABC

PERFORMING INFANT CPR

Danger: check for danger and maintain your own safety

Response

Gently shake the apparently unconscious child's shoulder and shout at her to elicit a response. If the child is conscious but has difficulty breathing, help her into a position that best eases her breathing—if she hasn't naturally assumed this position.

Send for help

Call for help to alert others and to enlist emergency assistance. If you're alone and the child isn't breathing, perform CPR for 1 minute before calling for help.

• Position the child in a supine position on a firm, flat surface (usually the ground). The surface should provide the resistance needed for adequate chest compressions. If you must turn the child from a prone position, support her head and neck and turn her as a unit to avoid injuring her spine (as shown below).



Airway

Kneel beside the child's shoulder. Place one hand on the child's forehead and gently lift her chin with your other hand to open her airway (as shown at top of next column). (In infants, this is called the sniffing position.) Avoid fingering the soft neck tissue to avoid obstructing the airway. Never let the child's mouth close completely.



• If you suspect a neck injury, use the jaw-thrust maneuver to open the child's airway to keep from moving the child's neck. To do this, kneel beside the child's head. With your elbows on the ground, rest your thumbs at the corners of the child's mouth, and place two or three fingers of each hand under the lower jaw. Lift the jaw upward.

Breathing

While maintaining an open airway, place your ear near the child's mouth and nose to evaluate her breathing status. Look for chest movement, listen for exhaled air, and feel for exhaled air on your cheek (as shown below).



• If the child is breathing, maintain an open airway and monitor respirations.

1-If they are breathing normally but they unconscious, the casualty should be rolled into the recovery position and send for help



2-If there is no breathing at all you should remove any obvious airway obstruction and give 5 initila rescues breaths, (deliver a gentle puff of air because an infant's lungs hold less air than an adult's. If the infant's chest rises and falls, the amount of air is probably adequate). After that check again for breathing if the casualty is unconscious and blue, compressions should be started.

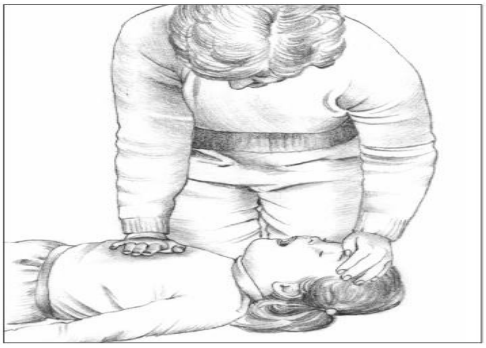
Compressions

• Assess circulation by palpating the carotid artery for a pulse in a child aged over 1 year.

•In an infant- feel for the brachial pulse or the inner aspect of the upper arm. If signs of life detected, rescuing of breathing should be continued until the child starts breathing effectively. If there are no signs of life start chest compressions

• Begin chest compressions if you find no pulse or if the pulse rate is less than 60 beats/minute with signs of poor perfusion. To locate the infant's heart, draw an imaginary line between the infant's nipples. Place three fingers directly below—and perpendicular to—the nipple line.

Then lift up your index finger so that the middle and ring fingers lie one fingerbreadth below the nipple line (as shown below). Use these two fingers to depress the sternum here ½″ to 1″ (1.3 to 2.5 cm) at least 100 compressions/minute. An alternative method for providing chest compressions is to place your thumbs side by side over the lower portion of the sternum and encircling the infant's chest, using the fingers of both hands.



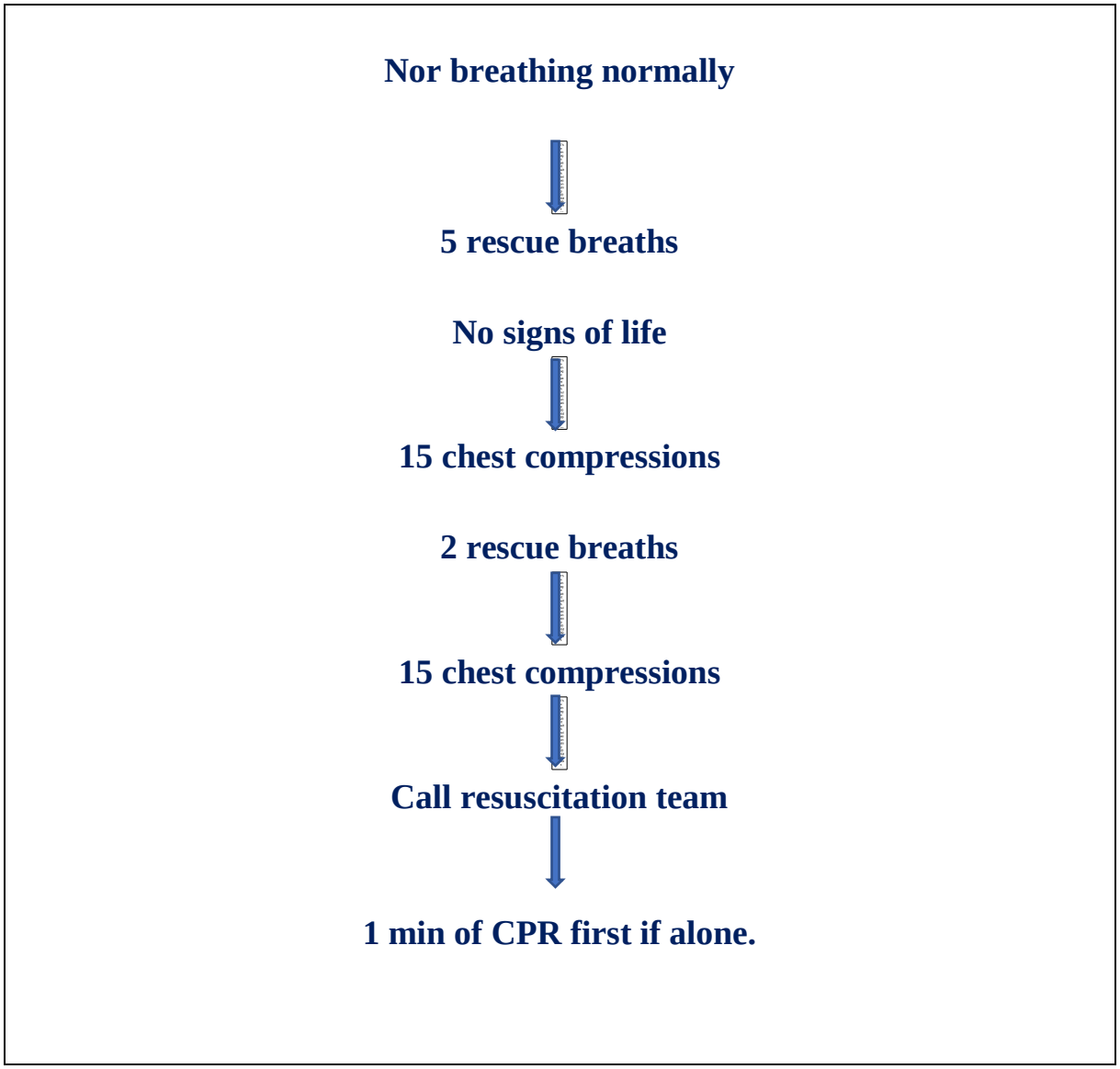
• Supply one breath after every five compressions. Maintain this ratio whether you're the helper or the lone rescuer. This ratio allows for about 100 compressions and 20 breaths/minute for an infant.

Note: If available, use a bag-valve mask over the infant's nose and mouth when performing ventilations.

Unresponsive

Shout for help

Open airway



Reference

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Dosage Calculation Problems

1. Order: Amoxil (amoxicillin) 70 mg PO q8h.
Patient: Child weighs 22 lbs.
Safe dose is 20–40 mg/kg/day in divided doses q8h.
Available as: 125 mg/5 mL
A. Is the dose safe? _____
B. If safe, how much liquid will you give the child per dose? _____
C. If not safe, what is the range of safe doses for this child? _____

2. Order: Phenobarbital 60 mg PO bid
Patient: Infant weighs 12 lbs 8 oz.
Safe dose is 4–8 mg/kg/day divided into two doses with Maximum dose of 300 mg/day.
Available as: Oral phenobarbital suspension of 30 mg/5 mL
A. Is the dose safe? _____
B. If safe, how much liquid will you give the child per dose? _____
C. If not safe, what is the range of safe doses for this child? _____

3. Order: Benadryl (diphenhydramine hydrochloride) 25 mg IV q6h prn
Patient: Child weighs 30 kg.
Safe dose is 1.25–10.5 mg/kg but not to exceed 200 mg/day.
Available as: Injection labeled 60 mg/mL
A. Is the dose safe? _____
B. If safe, how much will you draw up in the syringe for the child per dose? _____
C. If not safe, what is the safe dose for this child? _____

4. Order: Zithromax (azithromycin) 320 mg PO for one dose on first day

Patient: Child is 10 years old and weighs 32 kg.
Safe dose for children 2–15 years is 10 mg/kg (but not more than 500 mg/dose) on day 1.
Available as: Oral Zithromax suspension 100 mg/5 mL
A. Is the dose safe? _____
B. If safe, how much liquid will you give the child per dose? _____
C. If not safe, what is the safe dose for this child?_____

5. Order: Augmentin (amoxicillin clavulanate) 175 mg PO q8h.
Patient: Child weighs 29 lbs.
Safe dose is 40 mg/kg/day in divided doses q8h.
Available as: Oral suspension of 125 mg/5 mL
A. Is the dose safe? _____
B. If safe, how much liquid will you give the child per dose? _____
C. If not safe, what is the safe dose for this child? _____

6. Order: Lasix (furosemide) 15 mg PO bid
Patient: The child weighs 16 lbs 8 oz.
Safe dose for a child is 2 mg/kg/dose of furosemide as the initial dose.
Available as: 10 mg/mL
A. Is the dose safe?_____
B. If safe, how much in mL will you administer? _____

7. Order: Ampicillin 100 mg IV q 6 hrs
Patient: Child is 4 months old and weighs 9 lbs 6 oz.
Safe dose for this child's age and weight is 75–200 mg/kg/day.
Available as: Vial of powder 500 mg/vial.
Directions: Add 1.8 mL of sterile water; yields 250 mg/mL
A. What are the safe ranges? low range _____ high range_____
B. What is the safe dose?_____
C. How much will you give? _____

8. Order: Cefuroxime 185 mg IV q6h
Patient: Infant is 6 months old and weighs 17 lbs 6 oz.
Safe range is 50–100 mg/kg/day.
Available as: Cefuroxime 90 mg/mL
A. What are the safe ranges? low range _____
high range_____
B. Is the dose safe? _____
C. If safe, how much will you draw up?_____mL
9. Order: Epinephrine 0.3 mg sublingual
Patient: Child is 3 years old and weighs 66 lbs.
Safe dosage range is 0.01 mg/kg not to exceed 0.3 mg
for any single dose.
Available: Epinephrine 1 mg/mL
A. What are the safe ranges? low range _____
high range _____
B. Is the dose safe? _____
C. If safe, how many mL do you draw up for this
dose?_____
10. Order: Piperacillin 3375 mg IV q8h
Patient: Weighs 120 pounds.
Safe dosage range for adults is 4–8 grams
per day.
Available: Piperacillin 3375 mg in 100 mL of normal
saline to run over 30 minutes. You do not have a
pump and need to run the medication by drop factor.
You pick tubing that is 10 gtts/min.
A. Is the dose safe? _____
B. If safe, how many drops per minute will you regulate the tubing at
_____/minute?

ANSWER KEY

1. Dosage of Amoxil is safe at 210 mg/day or 70 mg per dose tid for a child weighing 22 lbs or 10 kg. Range is 200–400 mg per 24 hours. For 70-mg dose, give 2.8 mL of medication.

A. Safe—yes

B. 2.8 mL of medication

C. n/a

2. Dosage of Phenobarbital is not safe at 30 mg PO bid for infant weighing 12 lbs 8 oz or 5.68 kg. Range of doses at 4–8 mg per kg day divided into two doses is 22.72–45.44 mg per day or 11.36–22.72 mg per dose. You cannot give the medication.

A. Not safe—30 mg is too high (11.36–22.72 mg is the safe dose range)

B. No medication

C. 22.72–45.44 mg per day in two divided doses of 11.36–22.72 mg is safe range

3. Dosage of Benadryl is safe at 25 mg IV q6h for a child weighing 30 kg based on dose of 1.25–10.5 mg/kg/day, not to exceed 200 mg per day. Daily dosage is 100 mg and does not exceed 300 mg per day. With 60 mg per mL, the dose is 0.42 mL of medication to be drawn into syringe.

A. Safe—yes

B. 0.42 mL of medication

C. n/a

4. Dosage of Zithromax is safe at 320 mg PO for one dose for a child weighing 32 kg. Safe dosage is based on 10 mg per kg per dose. This is not more than 500 mg per day. With 100 mg per 5 mL of medication, you would give 16 mL of medication.

A. Safe—yes

B. 16 mL of medication
C. n/a
5. Dosage is safe at 125 mg for a child who weighs 29 pounds or 13.18 kg. Dosage is based on 40 mg per kg per day in 3 doses q8h. Weight of 13.18 kg times 40 mg/kg is 527.27 mg per day with 3 doses of 175 mg equals 525 mg per day. For 175-mg dose, give 7 mL of medication (125 mg per 5 mL).
A. Safe—yes
B. 7 mL of medication
C. n/a
6. Child weighs 7.5 kg; 15 mg/kg is safe dose.
A. Safe—yes
B. Divide 15 mg/10 mg × 1 mL = 1.5 mL
7. Child weighs 4.3 kg.
A. 75 × 4.3 kg = 322.5 mg/day
200 × 4.3 kg = 860 mg/day
B. 400 mg/day
C. 100 mg/500 mg × 1.8 mL = 0.4 mL
8. Child weighs 7.9 kg.
A. 50 × 7.9 kg = 395 mg/day 100 × 7.9 kg = 790 mg/day
B. Child is receiving 185 mg × 4 doses = 740 mg/day.
C. Safe—yes
D. 185 mg/90 mg × 1 mL = 2.1 mL
9. Child weighs 30 kg.
A. 0.01 mg × 30 kg = 0.3 mg/day
0.3 mg × 30 kg = 9 mg/day
B. Safe—yes
C. 0.3 mg/1 mg = 0.3 mL
10. Child weighs 54.5 kg.
A. No, it is not safe
B. None; I would not give.